

A COMMENTED ASSEMBLY LISTING
FOR THE
SHARP MZ80K
MONITOR

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New Bear Computer Store



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Prepared for Newbear
by Avalon Software
of England.

```
1 ; *****
2 ; * SHARP MZ-80K *
3 ; * MONITOR *
4 ; *****
5
6 ; Most of the code isn't available
7 ; from the command loop. The only
8 ; commands are:
9
10 ; LOAD...LOAD cassette
11 ; GOTO...GOTO user code
12 ; FD.....FLOPPY DISK
13 ; SG.....Keyboard beeps
14 ; SS.....Keyboard won't beep
15
16 ; HARDWARE.....
17
18 ; PORT0 is the decoder which
19 ; drives the keyboard rows
20
21 ; PORT1 is the input from the
22 ; keyboard column outputs.
23
24 ; PORT2 inputs are SENSE, READ,
25 ; CURSOR TIMER and VIDEOBLANK
26
27 ; PORT2 outputs are VGATE, WRITE,
28 ; LED and MOTOR.
29
30 ; PORT3 is the 8255 control port
31 ; for the above ports.
32
33 ; PORT4 is the 8253 timer 0, which
34 ; is the music divider.
35
36 ; PORT5 is the 8253 timer 1, which
37 ; is the CLOCK pre-scaler. It
38 ; divides the 31.250KHz signal
39 ; down to 1Hz.
40
41 ; PORT6 is the 8253 timer 2, which
42 ; is the CLOCK.
43
44 ; PORT7 is the 8253 control port
45
46 ; PORT8 is the music TEMPO timer
47 ; which is simply a 555.
48
49 ; VIDEO DISPLAY: this is a 40x 25
50 ; memory-mapped display. It uses a
51 ; non-ASCII character set. To
52 ; print an ASCII string you must
53 ; use routine SPECIAL to convert
54 ; from ASCII to the special
55 ; characters. Conversion is done
56 ; by simply looking up characters
57 ; in SPECTABLE.
58
59
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61      ; KEYBOARD: This is a software
62      ; scanned 10 row by 8 column.
63      ; matrix.
64
65      ; ----COLUMNS-----
66      ; 0011223344556677
67      ; ***** Rows 0,1
68      ; ***** Rows 2,3
69      ; ***** Rows 4,5
70      ; ***** Rows 6,7
71      ; ***** Rows 8,9
72
73      ; There are two electrical rows
74      ; of eight columns in one
75      ; physical row. Alternate keys
76      ; belongs to alternate rows. The
77      ; leftmost key of each column pair
78      ; belongs to the even-numbered
79      ; row. The rightmost key belongs to
80      ; the odd-numbered row.
81
82      ; To find the actual code prod-
83      ; uced by a key look up the row
84      ; and column in CAPTABLE.
85      ; For example key 'A' produces
86      ; 01H (Row 4, Column 0). Note that
87      ; the keyboard produces the same
88      ; non-ASCII characters as required
89      ; by the video display.
90      ; To convert to ASCII then
91      ; routine NORMAL uses NORMTABLE
92      ; in exactly the same way as
93      ; SPECIAL converts ASCII to the
94      ; special characters. Got it?
95
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121                                ORG    0
122
123      CR:      EQU    13          ; Carriage Return
124      ESCAPE:  EQU    27         ; Escape
125      BLANK:   EQU    32         ; Space

126      HALFDAY: EQU    43200      ; Secs in 12hrs.
127
128      VIDRAM:   EQU    0D000H     ; Video RAM
129      PORT0:    EQU    0E000H     ; Peripherals
130      PORT1:    EQU    0E001H
131      PORT2:    EQU    0E002H
132      PORT3:    EQU    0E003H
133      PORT4:    EQU    0E004H
134      PORT5:    EQU    0E005H
135      PORT6:    EQU    0E006H
136      PORT7:    EQU    0E007H
137      PORT8:    EQU    0E008H
138
139      ; Flags and buffers run from the
140      ; end of the monitor up to BASIC
141      ; at 1200H.
142
143      VECTOR:    EQU    1038H      ; Interrupt vector
144      STACK:     EQU    10F0H      ; Stack
145      HEADBUFFER: EQU    10F1H      ; File header
146      FILESIZE:  EQU    1102H      ; File size
147      FILESTART: EQU    1104H      ; Start address
148      EXECUTE:   EQU    1106H      ; File execute
149      CASEFLAG:  EQU    1170H      ; KBD Caps/Small
150      CURSOR:    EQU    1171H      ; Cursor pointer
151      VIDMAP:    EQU    1173H      ; Video flag map
152      VIDCHAR:   EQU    118EH      ; Hidden char
153      VIDPOS:    EQU    118FH      ; Point to above
154      ONOFF:     EQU    1191H      ; Cursor on/off
155      CURCHAR:   EQU    1192H      ; Char at cursor
156      USERMODE: EQU    1193H      ; User mode
157      CHARCOUNT: EQU    1194H      ; Counter
158      LEADCOUNT: EQU    1195H      ; Leader length
159      BITCOUNT: EQU    1197H      ; Checksum
160      BITCOUNT2: EQU    1199H
161      AMPM:      EQU    119BH      ; Clock AM/PM
162      CLOCKFLAG: EQU    119CH      ; Clock on
163      BEEPFLAG:  EQU    119DH      ; Beep on/off
164      TEMPO:     EQU    119EH      ; Music tempo
165      LENGTH:    EQU    119FH      ; Note length
166      OCTAVE:    EQU    11A0H      ; Music octave
167      FREQUENCY: EQU    11A1H      ; Note pitch
168      TEXTBUFFER: EQU    11A3H      ; Text
169
170      ; Code starts here with a table
171      ; of vectors.
172
173      0000 C34A00  COLDSTART: JP    INITIALISE      ; Reset
174      0003 C3E607  USER:      JP    User              ; User input handler.
175      0006 C30E09  CRLF2:   JP    Crlf2            ; Newline.
176      0009 C31809  CRLF:    JP    Crlf            ; Newline again..
177      000C C32009  SPACE:   JP    Space           ; Output a space.
178      000F C32609  TAB:     JP    Tab             ; Pad with spaces.
179      0012 C33509  VIDEO:   JP    Video           ; Video driver.
180      0015 C38109  MESSAGE: JP    Message         ; Output text.

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181	0018	C39909	STRING:	JP	String	; Output text.
182	001B	C3BD08	KEYBOARD:	JP	Keyboard	; KBD driver.
183	001E	C3320A	BREAK:	JP	Break	; Break test
184	0021	C33604	PHEAD:	JP	Phead	; Punch header.
185	0024	C37504	PDATA:	JP	Pdata	; Punch data.
186	0027	C3D804	LHEAD:	JP	Lhead	; Load header.
187	002A	C3F804	LDATA:	JP	Ldata	; Load data.
188	002D	C38805	CHECK:	JP	Check	; Checks file.
189	0030	C3C701	MUSIC:	JP	Music	; Music driver.
190	0033	C30803	SETCLOCK:	JP	Setclock	; Sets time.
191	0036	00		NOP		
192	0037	00		NOP		
193	0038	C33810	INTERRUPT:	JP	VECTOR	
194	003B	C35803	READTIME:	JP	Readtime	; Time
195	003E	C3E502	BEEP:	JP	Beep	; Keyboard beeper.
196	0041	C3FA02	SPEED:	JP	Speed	; Music routine.
197	0044	C3AB02	SOUND:	JP	Sound	; Music routine.
198	0047	C3BE02	SILENCE:	JP	Silence	; Music routine.
199						
200	004A	31F010	INITIALISE:	LD	SP,STACK	; Set stack
201	004D	ED56		IM	1	; Interrupts to 38H.
202	004F	CDC90F		CALL	INIT8255	; Init ports
203	0052	3E16		LD	A,16H	
204	0054	CD1200		CALL	VIDEO	; Clear screen.
205	0057	063C		LD	B,60	
206	0059	217011		LD	HL,CASEFLAG	
207	005C	CDD80F		CALL	SCRUB	; Zero all flags.
208	005F	219203		LD	HL,CLOCK	
209	0062	3EC3		LD	A,0C3H	; JUMP code
210	0064	323810		LD	(VECTOR),A	; Interrupts go
211	0067	228910		LD	(VECTOR+1),HL	; to clock
212	006A	3E04		LD	A,4	
213	006C	329E11		LD	(TEMPO),A	
214	006F	3C		INC	A	
215	0070	329F11		LD	(LENGTH),A	
216	0073	CDBE02		CALL	Silence	; Init music
217	0076	CD0900		CALL	CRLF	
218	0079	114101		LD	DE,M3	
219	007C	CD1500		CALL	MESSAGE	; Sign-on.
220	007F	C36B01		JP	SETS	; Go to mainloop.
221						
222	0082	CD0900	MAINLOOP:	CALL	CRLF	
223	0085	3E2A		LD	A,'*'	
224	0087	CD1200		CALL	VIDEO	; Prompt user.
225	008A	11A311		LD	DE,TEXTBUFFER	
226	008D	CD0300		CALL	USER	; Get a user line.
227	0090	1A		LD	A,(DE)	; Check first character.
228	0091	FE1B		CP	ESCAPE	; Abort?
229	0093	CA8200		JP	Z,MAINLOOP	
230	0096	FE2A		CP	'*'	
231	0098	2001		JR	NZ,MLP2	
232	009A	13		INC	DE	; Bump past nonsense.
233	009B	219601	MLP2:	LD	HL,M4	; Point 'LOAD'.
234	009E	0604		LD	B,4	; Text length.
235	00A0	CD8001		CALL	MATCH	; Check for match.
236	00A3	CACF00		JP	Z,LOADER	; Found match?
237	00A6	219A01		LD	HL,M5	; 'GOTO'
238	00A9	CD8001		CALL	MATCH	
239	00AC	CA5901		JP	Z,GOTO	; Match?
240	00AF	219E01		LD	HL,M6	; 'SS'

241	00B2	0602		LD	B,2	
242	00B4	CD8001		CALL	MATCH	
243	00B7	CA6B01		JP	Z,SETS	
244	00BA	21A001		LD	HL,M7	; 'SG'
245	00BD	CD8001		CALL	MATCH	
246	00C0	CA7301		JP	Z,SETG	
247	00C3	21A201		LD	HL,M8	; 'FD'
248	00C6	CD8001		CALL	MATCH	
249	00C9	CA7701		JP	Z,FDOS	
250	00CC	C38200		JP	MAINLOOP	; No match.
251						
252	00CF	13	LOADER:	INC	DE	; Bump past command.
253	00D0	13		INC	DE	
254	00D1	13		INC	DE	
255	00D2	13		INC	DE	
256	00D3	1A		LD	A,(DE)	; Pick up character.
257	00D4	FE0D		CP	CR	; Carriage return?
258	00D6	C20401		JP	NZ,FINDNAME	; If not, find file.
259	00D9	CDD804		CALL	Lhead	; Load file header.
260	00DC	DAA401		JP	C,BYE	; Abort flag.
261	00DF	CD0900	LD2:	CALL	CRLF	
262	00E2	113801		LD	DE,M2	; 'LOADING'
263	00E5	CD1500		CALL	MESSAGE	
264	00E8	11F110		LD	DE,HEADBUFFER	; Point header.
265	00EB	211000		LD	HL,16	
266	00EE	19		ADD	HL,DE	; Point end of name.
267	00EF	360D		LD	(HL),CR	; Terminate file name.
268	00F1	CD1500		CALL	MESSAGE	; Print file name.
269	00F4	CDF804		CALL	Ldata	; Load file data.
270	00F7	DAA401		JP	C,BYE	; Abort flag?
271	00FA	2A0611		LD	HL,(EXECUTE)	; Execute address.
272	00FD	7C		LD	A,H	; Pick up address MSB.
273	00FE	FE12		CP	12H	; Is it BASIC?
274	0100	DAB200		JP	C,MAINLOOP	; No.
275	0103	E9		JP	(HL)	; YES, so execute it.
276	0104	D5	FINDNAME:	PUSH	DE	; Save pointer.
277	0105	CDD804	FND2:	CALL	Lhead	; Load header.
278	0108	DAA401		JP	C,BYE	; Abort?
279	010B	CD0900		CALL	CRLF	
280	010E	113101		LD	DE,M1	; 'FOUND'
281	0111	CD1500		CALL	MESSAGE	
282	0114	11F110		LD	DE,HEADBUFFER	
283	0117	211000		LD	HL,16	
284	011A	19		ADD	HL,DE	
285	011B	360D		LD	(HL),CR	
286	011D	CD1500		CALL	MESSAGE	; Print file name.
287	0120	D1		POP	DE	; Pick up text pointer.
288	0121	D5		PUSH	DE	
289	0122	21F110		LD	HL,HEADBUFFER	; Point to name.
290	0125	0610		LD	B,16	; Name length.
291	0127	CD8001		CALL	MATCH	; The one we want?
292	012A	C20501		JP	NZ,FND2	; No, try again.
293	012D	D1		POP	DE	
294	012E	C3DF00		JP	LD2	; Yes, load it.
295						
296	0131	464F554E	M1:	DB	'FOUND ',CR	
296	0135	44200D				
297	0138	4C4F4144	M2:	DB	'LOADING ',CR	
297	013C	494E4720				
297	0140	0D				

298	0141	2A2A2020	M3:	DB	'** MONITOR '	
298	0145	4D4F4E49				
298	0149	544F5220				
299	014D	53502D31		DB	'SF-1002 **',CR	
299	0151	30303220				
299	0155	202A2A0D				
300						
301	0159	13	GOTO:	INC	DE	; Bump past command.
302	015A	13		INC	DE	
303	015B	13		INC	DE	
304	015C	13		INC	DE	
305	015D	1A		LD	A,(DE)	; Pick up character.
306	015E	FE24		CP	'\$'	; Check for nonsense.
307	0160	C28200		JP	NZ,MAINLOOP	
308	0163	13		INC	DE	; Bump to address.
309	0164	CD1004		CALL	CONVERT	; Convert to binary.
310	0167	DA8200		JP	C,MAINLOOP	; Bad address.
311	016A	E9		JP	(HL)	; Go do it.
312						
313	016B	3EFF	SETS:	LD	A,OFFH	; Beep off flas.
314	016D	329D11	SS2:	LD	(BEEPFLAG),A	
315	0170	C38200		JP	MAINLOOP	
316						
317	0173	AF	SETG:	XOR	A	; Beep on flas.
318	0174	C36D01		JP	SS2	
319						
320	0177	2100F0	FDOS:	LD	HL,OF000H	; Point to DOS
321	017A	7E		LD	A,(HL)	
322	017B	B7		OR	A	; DOS there?
323	017C	C28200		JP	NZ,MAINLOOP	
324	017F	E9		JP	(HL)	; Go to DOS.
325						
326	0180	C5	MATCH:	PUSH	BC	
327	0181	D5		PUSH	DE	
328	0182	E5		PUSH	HL	
329	0183	1A	MT2:	LD	A,(DE)	; Pick up text.
330	0184	BE		CP	(HL)	; Check against reference.
331	0185	200B		JR	NZ,MT3	; No match.
332	0187	05		DEC	B	
333	0188	2808		JR	Z,MT3	; Run out of characters.
334	018A	FE0D		CP	CR	; Shortened?
335	018C	2804		JR	Z,MT3	
336	018E	13		INC	DE	
337	018F	23		INC	HL	; Point to next.
338	0190	18F1		JR	MT2	; Keep going.
339	0192	E1	MT3:	POP	HL	
340	0193	D1		POP	DE	
341	0194	C1		POP	BC	
342	0195	C9		RET		
343						
344	0196	4C4F4144	M4:	DB	'LOAD'	
345	019A	474F544F	M5:	DB	'GOTO'	
346	019E	5353	M6:	DB	'SS'	
347	01A0	5347	M7:	DB	'SG'	
348	01A2	4644	M8:	DB	'FD'	
349						
350	01A4	FE02	BYE:	CP	2	; Was it a keypress?
351	01A6	CA8200		JP	Z,MAINLOOP	; Clean crashout.
352	01A9	CD0900		CALL	CRLF	
353	01AC	11E501		LD	DE,M9	

354	01AF	CD1500		CALL MESSAGE		; Announce tape error.
355	01B2	C38200		JP	MAINLOOP	
356						
357	01B5	22434845	M9:	DB	'CHECK SUM ERROR'',CR	
357	01B9	434B2053				
357	01BD	554D2045				
357	01C1	52524F52				
357	01C5	220D				
358						
359	01C7	C5	Music:	PUSH	BC	
360	01C8	D5		PUSH	DE	
361	01C9	E5		PUSH	HL	
362	01CA	3E02		LD	A,2	
363	01CC	32A011		LD	(OCTAVE),A	; Set middle octave.
364	01CF	0601		LD	B,1	; Initial tempo.
365	01D1	1A	MUS2:	LD	A,(DE)	; Pick up text.
366	01D2	FE0D		CP	CR	
367	01D4	2802		JR	Z,MUS3	
368	01D6	FEC8		CP	OC8H	
369	01D8	CA1002	MUS3:	JP	Z,MUSEXIT	; Time to finish?
370	01DB	FECF		CP	OCFH	
371	01DD	CA0302		JP	Z,LOWER	; Down an octave?
372	01E0	FED7		CP	OD7H	
373	01E2	CA0C02		JP	Z,HIGHER	; Up an octave?
374	01E5	FE23		CP	'E'	; Sharpened note?
375	01E7	217102		LD	HL,PITCHTAB	
376	01EA	2004		JR	NZ,MUS4	; Point normal notes
377	01EC	218902		LD	HL,SHARPTAB	; Sharp notes
378	01EF	13		INC	DE	; Bump past sharp sign.
379	01F0	CD1C02	MUS4:	CALL	PITCH	; Work out pitch.
380	01F3	DAD101		JP	C,MUS2	; Duff note.
381	01F6	CDC802		CALL	DURATION	; Work out length.
382	01F9	DA1302		JP	C,MUSEX2	
383	01FC	CDA802		CALL	Sound	; Make a noise
384	01FF	41		LD	B,C	; Pick up new tempo.
385	0200	C3D101		JP	MUS2	; Carry on.
386						
387	0203	3E03	LOWER:	LD	A,3	; Octave down.
388	0205	32A011	LOW2:	LD	(OCTAVE),A	
389	0208	13		INC	DE	
390	0209	C3D101		JP	MUS2	
391						
392	020C	3E01	HIGHER:	LD	A,1	; Octave up.
393	020E	18F5		JR	LOW2	
394						
395	0210	CDC802	MUSEXIT:	CALL	DURATION	; Note ends.
396	0213	F5	MUSEX2:	PUSH	AF	; Save BREAK flag.
397	0214	CDBE02		CALL	Silence	
398	0217	F1		POP	AF	
399	0218	E1		POP	HL	
400	0219	D1		POP	DE	
401	021A	C1		POP	BC	
402	021B	C9		RET		
403						
404	021C	C5	PITCH:	PUSH	BC	
405	021D	0608		LD	B,8	; Table length.
406	021F	1A		LD	A,(DE)	; Pick up text.
407	0220	BE	PIT2:	CP	(HL)	; Check table.
408	0221	2809		JR	Z,PIT3	; Got it?
409	0223	23		INC	HL	

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410 0224 23          INC HL
411 0225 23          INC HL          ; Bump to next entry.
412 0226 10F8        DJNZ PIT2      ; Keep going.
413 0228 37          SCF            ; Error flag.
414 0229 13          INC DE
415 022A C1          POP BC
416 022B C9          RET
417
418 022C 23          PIT3:          INC HL          ; Point to pitch.
419 022D D5          PUSH DE
420 022E 5E          LD E,(HL)
421 022F 23          INC HL
422 0230 56          LD D,(HL)      ; Pick up pitch value.
423 0231 EB          EX DE,HL
424 0232 7C          LD A,H
425 0233 B7          OR A          ; Zero is REST.
426 0234 2809        JR Z,PIT4
427 0236 3AA011      LD A,(OCTAVE)  ; Pick up octave.
428 0239 3D          PIT32:        DEC A
429 023A 2803        JR Z,PIT4
430 023C 29          ADD HL,HL      ; Drop by an octave.
431 023D 18FA        JR PIT32      ; Keep going.
432 023F 22A111      PIT4:          LD (FREQUENCY),HL ; Save it.
433 0242 3E02        LD A,2
434 0244 32A011      LD (OCTAVE),A  ; Reset octave.
435 0247 D1          POP DE
436 0248 13          INC DE
437 0249 1A          LD A,(DE)      ; Pick up text.
438 024A 47          LD B,A        ; Save length value.
439 024B E6F0        AND OFOH      ; Strip number.
440 024D EE30        CP '0'        ; Was it a digit?
441 024F 2805        JR Z,PIT5
442 0251 3A9F11      LD A,(LENGTH) ; Else pick default.
443 0254 1807        JR PIT6       ; And skip a bit.
444 0256 13          PIT5:          INC DE        ; Bump past length.
445 0257 78          LD A,B        ; Pick up length.
446 0258 E60F        AND OFH       ; Isolate digit.
447 025A 329F11      LD (LENGTH),A ; And save.
448 025D 4F          PIT6:          LD C,A
449 025E 0600        LD B,0
450 0260 21A102      LD HL,LENTAB  ; Point length table.
451 0263 09          ADD HL,BC      ; Point to entry.
452 0264 4E          LD C,(HL)
453 0265 3A9E11      LD A,(TEMPO)  ; Pick up tempo.
454 0268 47          LD B,A
455 0269 AF          XOR A
456 026A 81          PIT7:          ADD A,C        ; Double up length-
457 026B 10FD        DJNZ PIT7     ; According to tempo.
458 026D C1          POP BC
459 026E 4F          LD C,A        ; Save length.
460 026F AF          XOR A        ; OK flag.
461 0270 C9          RET
462
463                  ; The pitch tables consist of a
464                  ; key letter and a pitch value. The
465                  ; pitch value is loaded
466                  ; into Timer 0 of the 8253 timer
467                  ; as a division integer.
468
469 0271 43          PITCHTAB: DB 'C'

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470	0272	7707		DW	1911	; 523HZ
471	0274	44		DB	'D'	
472	0275	A706		DW	1703	; 587HZ
473	0277	45		DB	'E'	
474	0278	ED05		DW	1517	; 659HZ
475	027A	46		DB	'F'	
476	027B	9805		DW	1432	; 698HZ
477	027D	47		DB	'G'	
478	027E	FC04		DW	1276	; 784HZ
479	0280	41		DB	'A'	
480	0281	7104		DW	1137	; 880HZ
481	0283	42		DB	'B'	
482	0284	F503		DW	1013	; 987HZ
483	0286	52		DB	'R'	
484	0287	0000		DW	0	; REST
485						
486	0289	43	SHARFTAB:	DB	'C'	
487	028A	QC07		DW	1804	; 554HZ
488	028C	44		DB	'D'	
489	028D	4706		DW	1607	; 622HZ
490	028F	45		DB	'E'	
491	0290	9805		DW	1432	; 698HZ
492	0292	46		DB	'F'	
493	0293	4805		DW	1352	; 740HZ
494	0295	47		DB	'G'	
495	0296	B404		DW	1204	; 831HZ
496	0298	41		DB	'A'	
497	0299	3104		DW	1073	; 932HZ
498	029B	42		DB	'B'	
499	029C	BB03		DW	955	; 1047HZ
500	029E	52		DB	'R'	
501	029F	0000		DW	0	; REST
502						
503	02A1	01020304	LENTAB:	DB	1,2,3,4	
504	02A5	06080C10		DB	6,8,12,16	
505	02A9	1820		DB	24,32	
506						
507	02AB	2AA111	Sound:	LD	HL,(FREQUENCY)	
508	02AE	7C	SOUND1:	LD	A,H	
509	02AF	B7		OR	A	; Rest?
510	02B0	280C		JR	Z,Silence	
511	02B2	D5		PUSH	DE	
512	02B3	EB		EX	DE,HL	
513	02B4	2104E0		LD	HL,PORT4	; 8253 Timer 0.
514	02B7	73		LD	(HL),E	; Pitch LSB.
515	02B8	72		LD	(HL),D	; Pitch MSB.
516	02B9	3E01		LD	A,1	
517	02BB	D1		POP	DE	
518	02BC	1806		JR	SIL2	; Activate 555 tempo timer
519						
520	02BE	3E34	Silence:	LD	A,34H	; Mode 2 Binary-
521	02C0	3207E0		LD	(PORT7),A	; Access timer twice.
522	02C3	AF		XOR	A	; Reset 555 tempo timer.
523	02C4	3208E0	SIL2:	LD	(PORT8),A	; Tempo timer.
524	02C7	C9		RET		
525						
526	02C8	2100E0	DURATION:	LD	HL,PORT0	
527	02CB	36F9		LD	(HL),0F9H	; KBD row 9
528	02CD	23		INC	HL	
529	02CE	7E		LD	A,(HL)	; Look at columns

530	02DF	E60E		AND	B		; Isolate BREAK key.
531	02D1	2002		JR	NZ, DUR2		
532	02D3	37		SCF			; BREAK pressed.
533	02D4	C9		RET			
534	02D5	3A08E0	DUR2:	LD	A, (PORT8)		
535	02D8	0F		RRCA			
536	02D9	38FA		JR	C, DUR2		; Wait for tempo pulse.
537	02DB	3A08E0	DUR3:	LD	A, (PORT8)		
538	02DE	0F		RRCA			
539	02DF	30FA		JR	NC, DUR3		
540	02E1	10F2		DJNZ	DUR2		; Keep going.
541	02E3	AF		XOR	A		
542	02E4	C9		RET			
543							
544	02E5	C5	Beep:	PUSH	BC		
545	02E6	E5		PUSH	HL		
546	02E7	217104		LD	HL, 1137		; Pitch
547	02EA	CDAE02		CALL	SOUND1		; Make a beep!
548	02ED	0632		LD	B, 50		
549	02EF	AF	BP2:	XOR	A		
550	02F0	CD5B07		CALL	DEL1		; Waste time.
551	02F3	10FA		DJNZ	BP2		
552	02F5	E1		POP	HL		
553	02F6	C1		POP	BC		
554	02F7	C3BE02		JP	Silence		; Peace
555							
556	02FA	F5	Speed:	PUSH	AF		
557	02FB	C5		PUSH	BC		
558	02FC	E60F		AND	0FH		; Isolate digit.
559	02FE	47		LD	B, A		
560	02FF	3E08		LD	A, 8		
561	0301	90		SUB	B		
562	0302	329E11		LD	(TEMPO), A		
563	0305	C1		POP	BC		
564	0306	F1		POP	AF		
565	0307	C9		RET			
566							
567	0308	F3	Setclock:	DI			; Interrupts off.
568	0309	C5		PUSH	BC		
569	030A	D5		PUSH	DE		
570	030B	E5		PUSH	HL		
571	030C	329B11		LD	(AMPM), A		; Set AM/PM
572	030F	3EFO		LD	A, 0FOH		
573	0311	329C11		LD	(CLOCKFLAG), A		; Clock is running.
574	0314	21C0AB		LD	HL, HALFDAY		
575	0317	AF		XOR	A		
576	0318	ED52		SBC	HL, DE		; Clock counts backwards!
577	031A	E5		PUSH	HL		
578	031B	23		INC	HL		
579	031C	EB		EX	DE, HL		
580	031D	3E74		LD	A, 74H		; Mode 2 Binary-access
581	031F	3207E0		LD	(PORT7), A		; twice, Timer 1.
582	0322	3EB0		LD	A, 0BOH		; Mode 0 Binary access-
583	0324	3207E0		LD	(PORT7), A		; twice, Timer 2.
584	0327	2106E0		LD	HL, PORT6		
585	032A	73		LD	(HL), E		
586	032B	72		LD	(HL), D		; Set the time.
587	032C	2B		DEC	HL		; Point Timer 1, divider.
588	032D	360A		LD	(HL), 10		; Divide by ten, clock-
589	032F	3600		LD	(HL), 0		; runs 3125 times as fast.

590	0331	3E80		LD	A, 80H	; Prepare to read time.
591	0333	3207E0		LD	(PORT7), A	
592	0336	23		INC	HL	
593	0337	4E	SETCL2:	LD	C, (HL)	; Read time.
594	0338	7E		LD	A, (HL)	
595	0339	BA		CP	D	; Right time yet?
596	033A	20FB		JR	NZ, SETCL2	
597	033C	79		LD	A, C	
598	033D	BB		CP	E	
599	033E	20F7		JR	NZ, SETCL2	; Keep going.
600	0340	2B		DEC	HL	; Point Timer 1
601	0341	00		NOP		
602	0342	00		NOP		
603	0343	00		NOP		
604	0344	3612		LD	(HL), 31250 & 255	; Divide LSB.
605	0346	367A		LD	(HL), 31250/256	; Divide MSB.
606	0348	23		INC	HL	
607	0349	D1		POP	DE	
608	034A	4E	SETCL3:	LD	C, (HL)	
609	034B	7E		LD	A, (HL)	; Read time again.
610	034C	BA		CP	D	
611	034D	20FB		JR	NZ, SETCL3	
612	034F	79		LD	A, C	
613	0350	BB		CP	E	; at normal speed.
614	0351	20F7		JR	NZ, SETCL3	
615	0353	E1		POP	HL	
616	0354	D1		POP	DE	
617	0355	C1		POP	BC	
618	0356	FB		EI		; Interrupts back on.
619	0357	C9		RET		
620						
621	0358	E5	Readtime:	PUSH	HL	
622	0359	3E80		LD	A, 80H	
623	035B	3207E0		LD	(PORT7), A	; Timer 2.
624	035E	2106E0		LD	HL, PORT6	; Prepare to read.
625	0361	F3		DI		
626	0362	5E		LD	E, (HL)	
627	0363	56		LD	D, (HL)	
628	0364	FB		EI		
629	0365	7B		LD	A, E	
630	0366	B2		OR	D	; Timed out?
631	0367	CA7903		JP	Z, RT2	
632	036A	AF		XOR	A	
633	036B	21COA8		LD	HL, HALFDAY	
634	036E	ED52		SBC	HL, DE	; Running backwards.
635	0370	DA8303		JP	C, RT4	; Overflowed?
636	0373	EB		EX	DE, HL	
637	0374	3A9B11		LD	A, (AMPM)	
638	0377	E1		POP	HL	
639	0378	C9		RET		
640	0379	11COA8	RT2:	LD	DE, HALFDAY	; Reset
641	037C	3A9B11	RT3:	LD	A, (AMPM)	
642	037F	EE01		XOR	1	; Flip the flag.
643	0381	E1		POP	HL	
644	0382	C9		RET		
645	0383	F3	RT4:	DI		
646	0384	2106E0		LD	HL, PORT6	
647	0387	7E		LD	A, (HL)	
648	0388	2F		CPL		; Form complement to find-
649	0389	5F		LD	E, A	; how far we've over-

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650 038A 7E          LD  A,(HL)          ; shot time-out.
651 038B 2F          CPL
652 038C 57          LD  D,A
653 038D FB          EI
654 038E 13          INC  DE
655 038F C37C03      JF   RT3            ; Exit.
656
657                  ; This is the interrupt-driven
658                  ; time keeping routine.
659                  ; Interrupts occur once every
660                  ; halfday (12hours or 43,200sec).
661
662 0392 F5          CLOCK:  PUSH AF
663 0393 C5          PUSH BC
664 0394 D5          PUSH DE
665 0395 E5          PUSH HL
666 0396 3A9B11      LD   A,(AMPM)
667 0397 EE01        XOR   1              ; Flip mornings/afternoon.
668 0398 329B11      LD   (AMPM),A
669 0399 3E80        LD   A,80H
670 03A0 3207E0      LD   (PORT7),A
671 03A1 2106E0      LD   HL,PORT6
672 03A2 5E          LD   E,(HL)
673 03A3 56          LD   D,(HL)
674 03A4 21C0A8      LD   HL,HALFDAY
675 03A5 19          ADD  HL,DE
676 03A6 2B          DEC  HL
677 03A7 2B          DEC  HL
678 03A8 EB          EX   DE,HL
679 03A9 2106E0      LD   HL,PORT6
680 03AA 73          LD   (HL),E
681 03AB 72          LD   (HL),D          ; Reset time.
682 03AC E1          POP  HL
683 03AD D1          POP  DE
684 03AE C1          POP  BC
685 03AF F1          POP  AF
686 03B0 FB          EI
687 03B1 C9          RET
688
689 03BA 7C          WORD:  LD   A,H          ; Pick up MSB.
690 03BB CDC303      CALL BYTE           ; Print in hex.
691 03BC 7D          LD   A,L          ; Pick up LSB.
692 03BD CDC303      CALL BYTE           ; Print in hex.
693 03BE C9          RET
694
695 03C3 F5          BYTE:  PUSH AF
696 03C4 E6F0        AND  OF0H          ; Isolate top nibble.
697 03C5 0F          RRCA              ; Shift to lower nibble.
698 03C6 0F          RRCA
699 03C7 0F          RRCA
700 03C8 0F          RRCA
701 03C9 0F          RRCA
702 03CA CDDA03      CALL NIBBLE         ; Convert to ASCII.
703 03CB CD1200      CALL VIDEO         ; Display it.
704 03CC F1          POP  AF
705 03CD E60F        AND  0FH          ; Lower nibble.
706 03CE CDDA03      CALL NIBBLE
707 03CF CD1200      CALL VIDEO
708 03D0 C9          RET
709 03DA D5          NIBBLE:  PUSH DE

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710	03DB	E5		PUSH	HL	
711	03DC	21E903		LD	HL, ASCIIITAB	; Point ASCII.
712	03DF	E60F		AND	0FH	
713	03E1	5F		LD	E, A	
714	03E2	1600		LD	D, 0	
715	03E4	19		ADD	HL, DE	; Point to entry.
716	03E5	7E		LD	A, (HL)	; Pick up ASCII.
717	03E6	E1		POP	HL	
718	03E7	D1		POP	DE	
719	03E8	C9		RET		
720						
721	03E9	30313233	ASCIIITAB:	DB	'0123'	
722	03ED	34353637		DB	'4567'	
723	03F1	38394142		DB	'89AB'	
724	03F5	43444546		DB	'CDEF'	
725						
726	03F9	C5	GETNIBBLE:	PUSH	BC	
727	03FA	E5		PUSH	HL	
728	03FB	010010		LD	BC, 16*256	; Table length
729	03FE	21E903		LD	HL, ASCIIITAB	
730	0401	BE	GTN2:	CP	(HL)	; Match?
731	0402	2003		JR	NZ, GTN3	
732	0404	79		LD	A, C	; Pick up table index.
733	0405	1806		JR	GTN4	; Finish
734	0407	23	GTN3:	INC	HL	
735	0408	0C		INC	C	
736	0409	05		DEC	B	
737	040A	20F5		JR	NZ, GTN2	
738	040C	37		SCF		; Error flag.
739	040D	E1	GTN4:	POP	HL	
740	040E	C1		POP	BC	
741	040F	C9		RET		
742						
743	0410	D5	CONVERT:	PUSH	DE	
744	0411	CD1F04		CALL	GETBYTE	; Get MSB
745	0414	3807		JR	C, CON2	; Error.
746	0416	67		LD	H, A	
747	0417	CD1F04		CALL	GETBYTE	; LSB.
748	041A	3801		JR	C, CON2	
749	041C	6F		LD	L, A	
750	041D	D1	CON2:	POP	DE	
751	041E	C9		RET		; HL is binary.
752						
753	041F	C5	GETBYTE:	PUSH	BC	
754	0420	1A		LD	A, (DE)	; Pick up ASCII text.
755	0421	13		INC	DE	; Bump pointer.
756	0422	CDF903		CALL	GETNIBBLE	
757	0425	380D		JR	C, GBY2	
758	0427	07		RLCA		
759	0428	07		RLCA		
760	0429	07		RLCA		
761	042A	07		RLCA		
762	042B	4F		LD	C, A	; Save top nibble.
763	042C	1A		LD	A, (DE)	
764	042D	13		INC	DE	
765	042E	CDF903		CALL	GETNIBBLE	
766	0431	3801		JR	C, GBY2	
767	0433	B1		OR	C	; Mix in the two nibbles.
768	0434	C1	GBY2:	POP	BC	
769	0435	C9		RET		

770			
771	0436	F3	Phead:
772	0437	D5	
773	0438	C5	
774	0439	E5	
775	043A	16D7	
776	043C	1ECC	
777	043E	21F010	
778	0441	018000	
779	0444	CD3307	PHD1:
780	0447	CDB206	
781	044A	DA6305	
782	044D	7B	
783	044E	FECC	
784	0450	2011	
785	0452	CD0900	
786	0455	D5	
787	0456	116C04	
788	0459	CD1500	
789	045C	11F110	
790	045F	CD1500	
791	0462	D1	
792	0463	CDB807	PHD2:
793	0466	CD8D04	
794	0469	C36305	
795			
796	046C	57524954	M10:
796	0470	494E4720	
796	0474	0D	
797			
798	0475	F3	Pdata:
799	0476	D5	
800	0477	C5	
801	0478	E5	
802	0479	16D7	
803	047B	1E53	
804	047D	2A0211	
805	0480	E5	
806	0481	C1	
807	0482	2A0411	
808	0485	7B	
809	0486	B1	
810	0487	CAD404	
811	048A	C34404	
812			
813	048D	D5	PUNCH:
814	048E	C5	
815	048F	E5	
816	0490	1602	
817	0492	3EF9	
818	0494	3200E0	
819	0497	7E	PUN1:
820	0498	CDA507	
821	049B	3A01E0	
822	049E	E608	
823	04A0	C2A704	
824	04A3	37	
825	04A4	C3D404	
826	04A7	23	PUN2:
827	04A8	0B	

828	04A9	78		LD	A,B	
829	04AA	B1		OR	C	; None?
830	04AB	C29704		JP	NZ,PUN1	
831	04AE	2A9711	PUN3:	LD	HL,(BITCOUNT)	
832	04B1	7C		LD	A,H	
833	04B2	CDA507		CALL	SRLOUT	; Punch checksum.
834	04B5	7D		LD	A,L	
835	04B6	CDA507		CALL	SRLOUT	
836	04B9	CD8007		CALL	LONGBIT	; Trailer.
837	04BC	15		DEC	D	
838	04BD	C2C404		JP	NZ,PUN4	; Punch twice.
839	04C0	B7		OR	A	
840	04C1	C3D404		JP	POPPER2	
841	04C4	0600	PUN4:	LD	B,0	
842	04C6	CD6707	PUN5:	CALL	SHORTBIT	; Trailer.
843	04C9	05		DEC	B	
844	04CA	C2C604		JP	NZ,PUN5	
845	04CD	E1		POP	HL	
846	04CE	C1		POP	BC	
847	04CF	C5		PUSH	BC	
848	04D0	E5		PUSH	HL	; Pick up parameters.
849	04D1	C39704		JP	PUN1	; Do it all again.
850						
851	04D4	E1	POPPER2:	POP	HL	
852	04D5	C1		POP	BC	
853	04D6	D1		POP	DE	
854	04D7	C9		RET		
855						
856	04D8	F3	Lhead:	DI		
857	04D9	D5		PUSH	DE	
858	04DA	C5		PUSH	BC	
859	04DB	E5		PUSH	HL	
860	04DC	16D2		LD	D,0D2H	; Load flas.
861	04DE	1ECC		LD	E,0CCH	; Header flas.
862	04E0	018000		LD	BC,128	
863	04E3	21F010		LD	HL,HEADBUFFER-1	
864	04E6	CDB206	LDH1:	CALL	STARTUP	; Motor on.
865	04E9	DA8205		JP	C,ABORT	
866	04EC	CD5E06		CALL	LLEADER	; Find leader.
867	04EF	DA8205		JP	C,ABORT	
868	04F2	CD1005		CALL	LOADIT	; Load stuff.
869	04F5	C36305		JP	TAPEXIT	; Finish.
870						
871	04F8	F3	Ldata:	DI		
872	04F9	D5		PUSH	DE	
873	04FA	C5		PUSH	BC	
874	04FB	E5		PUSH	HL	
875	04FC	16D2		LD	D,0D2H	; Load flas.
876	04FE	1E53		LD	E,53H	; Data flas.
877	0500	2A0211		LD	HL,(FILESIZE)	
878	0503	E5		PUSH	HL	
879	0504	C1		POP	BC	
880	0505	2A0411		LD	HL,(FILESTART)	
881	0508	78		LD	A,B	
882	0509	B1		OR	C	; Null file size?
883	050A	CA6305		JP	Z,TAPEXIT	
884	050D	C3E604		JP	LDH1	; Go load data.
885						
886	0510	D5	LOADIT:	PUSH	DE	
887	0511	C5		PUSH	BC	

888	0512	E5		PUSH HL	
889	0513	2602		LD H,2	
890	0515	0101E0	LDT1:	LD BC,PORT1	; Keyboard.
891	0518	1102E0		LD DE,PORT2	; Read input.
892	051B	CD0106	LDT2:	CALL EDGE	; Find an edge.
893	051E	DA8205		JP C,ABORT	
894	0521	CD6007		CALL DELAY2	; Wait to sample.
895	0524	CD6007		CALL DELAY2	
896	0527	CD6007		CALL DELAY2	
897	052A	1A		LD A,(DE)	; Sample.
898	052B	E620		AND 20H	; Isolate read line.
899	052D	CA1B05		JP Z,LDT2	; Wait for long bit.
900	0530	54		LD D,H	
901	0531	210000		LD HL,0	
902	0534	229711		LD (BITCOUNT),HL	; Zero checksum.
903	0537	E1		POP HL	
904	0538	C1		POP BC	
905	0539	C5		PUSH BC	
906	053A	E5		PUSH HL	; Pick up parameters.
907	053B	CD2406	LDT3:	CALL SRLIN	; Read a byte.
908	053E	DA8205		JP C,ABORT	
909	0541	77		LD (HL),A	
910	0542	23		INC HL	
911	0543	0B		DEC BC	
912	0544	78		LD A,B	
913	0545	B1		OR C	
914	0546	C23B05		JP NZ,LDT3	; Keep going.
915	0549	2A9711		LD HL,(BITCOUNT)	
916	054C	CD2406		CALL SRLIN	
917	054F	DA8205		JP C,ABORT	
918	0552	5F		LD E,A	; Tape checksum.
919	0553	CD2406		CALL SRLIN	
920	0556	DA8205		JP C,ABORT	
921	0559	BD		CP L	; Compare checksums.
922	055A	C27405		JP NZ,TRYAGAIN	
923	055D	7B		LD A,E	
924	055E	BC		CP H	
925	055F	C27405		JP NZ,TRYAGAIN	
926	0562	AF	GOODNEWS:	XOR A	; OK!
927	0563	E1	TAPEXIT:	POP HL	
928	0564	C1		POP BC	
929	0565	D1		POP DE	
930	0566	CD0007		CALL MOTOROFF	
931	0569	F5		PUSH AF	
932	056A	3A9C11		LD A,(CLOCKFLAG)	; Clock on?
933	056D	FEF0		CP OF0H	
934	056F	2001		JR NZ,TAPEX2	
935	0571	FB		EI	; Leave interrupts on.
936	0572	F1	TAPEX2:	POP AF	
937	0573	C9		RET	
938					
939	0574	15	TRYAGAIN:	DEC D	; First try?
940	0575	CA7C05		JP Z,BADNEWS	
941	0578	62		LD H,D	; Save pass count.
942	0579	C31505		JP LDT1	; Try second copy.
943					
944	057C	3E01	BADNEWS:	LD A,1	; Tape error flag.
945	057E	37		SCF	; Abort flag.
946	057F	C36305		JP TAPEXIT	
947					

948	0582	3E02	ABORT:	LD	A,2	; Abort flag.
949	0584	37		SCF		
950	0585	C36305		JP	TAPEXIT	
951						
952	0588	F3	Check:	DI		; Check & CHECKIT are the
953	0589	D5		PUSH	DE	; same as Ldata and LOADIT
954	058A	C5		PUSH	BC	; but data is not loaded,
955	058B	E5		PUSH	HL	; just compared to tape.
956	058C	2A0211		LD	HL,(FILESIZE)	
957	058F	E5		PUSH	HL	
958	0590	C1		POP	BC	
959	0591	2A0411		LD	HL,(FILESTART)	
960	0594	16D2		LD	D,0D2H	
961	0596	1E53		LD	E,53H	
962	0598	78		LD	A,B	
963	0599	B1		OR	C	
964	059A	CA6305		JP	Z,TAPEXIT	
965	059D	CD3307		CALL	COUNTRITS	
966	05A0	CDB206		CALL	STARTUP	
967	05A3	DA8205		JP	C,ABORT	
968	05A6	CD5E06		CALL	LLEADER	
969	05A9	DA8205		JP	C,ABORT	
970	05AC	CDB205		CALL	CHECKIT	
971	05AF	C36305		JP	TAPEXIT	
972						
973	05B2	D5	CHECKIT:	PUSH	DE	
974	05B3	C5		PUSH	BC	
975	05B4	E5		PUSH	HL	
976	05B5	2602		LD	H,2	
977	05B7	0101E0	CHK1:	LD	BC,PORT1	
978	05BA	1102E0		LD	DE,PORT2	
979	05BD	CD0106	CHK2:	CALL	EDGE	
980	05C0	DA8205		JP	C,ABORT	
981	05C3	CD6007		CALL	DELAY2	
982	05C6	CD6007		CALL	DELAY2	
983	05C9	CD6007		CALL	DELAY2	
984	05CC	1A		LD	A,(DE)	
985	05CD	E620		AND	20H	
986	05CF	CABD05		JP	Z,CHK2	
987	05D2	54		LD	D,H	
988	05D3	E1		POP	HL	
989	05D4	C1		POP	BC	
990	05D5	C5		PUSH	BC	
991	05D6	E5		PUSH	HL	
992	05D7	CD2406	CHK3:	CALL	SRLIN	
993	05DA	DA8205		JP	C,ABORT	
994	05DD	BE		CP	(HL)	
995	05DE	C27C05		JP	NZ,BADNEWS	
996	05E1	23		INC	HL	
997	05E2	0B		DEC	BC	
998	05E3	78		LD	A,B	
999	05E4	B1		OR	C	
1000	05E5	C2D705		JP	NZ,CHK3	
1001	05E8	2A9911		LD	HL,(BITCOUNT2)	
1002	05EB	CD2406		CALL	SRLIN	
1003	05EE	BC		CP	H	
1004	05EF	C27C05		JP	NZ,BADNEWS	
1005	05F2	CD2406		CALL	SRLIN	
1006	05F5	BD		CP	L	
1007	05F6	C27C05		JP	NZ,BADNEWS	

1008	05F9	15		DEC	D	
1009	05FA	CA6205		JP	Z,GOODNEWS	
1010	05FD	62		LD	H,D	
1011	05FE	C3B705		JP	CHK1	
1012						
1013	0601	3EF9	EDGE:	LD	A,0F9H	
1014	0603	3200E0		LD	(PORT0),A	; KBD row 9.
1015	0606	00		NOP		
1016	0607	0A	EDG1:	LD	A,(BC)	; Look at columns.
1017	0608	E608		AND	8	; Isolate BREAK.
1018	060A	C20F06		JP	NZ,EDG2	
1019	060D	37		SCF		
1020	060E	C9		RET		
1021	060F	1A	EDG2:	LD	A,(DE)	
1022	0610	E620		AND	20H	; Look at read line.
1023	0612	C20706		JP	NZ,EDG1	
1024	0615	0A	EDG3:	LD	A,(BC)	
1025	0616	E608		AND	8	
1026	0618	C21D06		JP	NZ,EDG4	
1027	061B	37		SCF		
1028	061C	C9		RET		
1029	061D	1A	EDG4:	LD	A,(DE)	
1030	061E	E620		AND	20H	
1031	0620	CA1506		JP	Z,EDG3	; Wait for leading edge.
1032	0623	C9		RET		
1033						
1034	0624	C5	SRLIN:	PUSH	BC	
1035	0625	D5		PUSH	DE	
1036	0626	E5		PUSH	HL	
1037	0627	210008		LD	HL,8*256	; 8bits per byte.
1038	062A	0101E0		LD	BC,PORT1	
1039	062D	1102E0		LD	DE,PORT2	
1040	0630	CD0106	SLN2:	CALL	EDGE	
1041	0633	DA5A06		JP	C,SLN4	; Abort.
1042	0636	CD6007		CALL	DELAY2	
1043	0639	CD6007		CALL	DELAY2	
1044	063C	CD6007		CALL	DELAY2	
1045	063F	1A		LD	A,(DE)	; Sample.
1046	0640	E620		AND	20H	; Isolate read line.
1047	0642	CA4F06		JP	Z,SLN3	; Zero = ZERO BIT
1048	0645	E5		PUSH	HL	
1049	0646	2A9711		LD	HL,(BITCOUNT)	
1050	0649	23		INC	HL	; ONE so bump checksum.
1051	064A	229711		LD	(BITCOUNT),HL	
1052	064D	E1		POP	HL	
1053	064E	37		SCF		; Shift a ONE into result
1054	064F	7D	SLN3:	LD	A,L	; Pick up result byte.
1055	0650	17		RLA		; Shift in bit.
1056	0651	6F		LD	L,A	
1057	0652	25		DEC	H	; More bits?
1058	0653	C23006		JP	NZ,SLN2	
1059	0656	CD0106		CALL	EDGE	; Sync.
1060	0659	7D		LD	A,L	; Pick up byte.
1061	065A	E1	SLN4:	POP	HL	
1062	065B	D1		POP	DE	
1063	065C	C1		POP	BC	
1064	065D	C9		RET		
1065						
1066	065E	C5	LLEADER:	PUSH	BC	
1067	065F	D5		PUSH	DE	

1068	0660	E5		PUSH	HL	
1069	0661	212828		LD	HL, 40*256+40	
1070	0664	7B		LD	A, E	
1071	0665	FECC		CP	OCCH	; Header flag?
1072	0667	CA6D06		JP	Z, LLEAD2	
1073	066A	211414		LD	HL, 20*256+20	
1074	066D	229511	LLEAD2:	LD	(LEADCOUNT), HL	
1075	0670	0101E0		LD	BC, PORT1	
1076	0673	1102E0		LD	DE, PORT2	
1077	0676	2A9511	LLEAD3:	LD	HL, (LEADCOUNT)	
1078	0679	CD0106	LLEAD4:	CALL	EDGE	
1079	067C	DAAE06		JP	C, LLEAD6	
1080	067F	CD6007		CALL	DELAY2	
1081	0682	CD6007		CALL	DELAY2	
1082	0685	CD6007		CALL	DELAY2	
1083	0688	1A		LD	A, (DE)	
1084	0689	E620		AND	20H	
1085	068B	CA7606		JP	Z, LLEAD3	; Find a long pulse.
1086	068E	25		DEC	H	
1087	068F	C27906		JP	NZ, LLEAD4	; Find 40 or 20.
1088	0692	CD0106	LLEAD5:	CALL	EDGE	
1089	0695	DAAE06		JP	C, LLEAD6	
1090	0698	CD6007		CALL	DELAY2	
1091	069B	CD6007		CALL	DELAY2	
1092	069E	CD6007		CALL	DELAY2	
1093	06A1	1A		LD	A, (DE)	
1094	06A2	E620		AND	20H	
1095	06A4	C27606		JP	NZ, LLEAD3	; Find short pulse.
1096	06A7	2D		DEC	L	
1097	06AB	C29206		JP	NZ, LLEAD5	; Find 40 or 20.
1098	06AB	CD0106		CALL	EDGE	
1099	06AE	E1	LLEAD6:	POP	HL	
1100	06AF	D1		POP	DE	
1101	06B0	C1		POP	BC	
1102	06B1	C9		RET		
1103						
1104	06B2	C5	STARTUP:	PUSH	BC	
1105	06B3	D5		PUSH	DE	
1106	06B4	E5		PUSH	HL	
1107	06B5	0EOA		LD	C, 10	
1108	06B7	3A02E0	STUP2:	LD	A, (PORT2)	
1109	06BA	E610		AND	10H	; Isolate motor sense.
1110	06BC	CAC406		JP	Z, STUP5	
1111	06BF	AF	STUP3:	XOR	A	
1112	06C0	E1	STUP4:	POP	HL	; Motor's on.
1113	06C1	D1		POP	DE	
1114	06C2	C1		POP	BC	
1115	06C3	C9		RET		
1116	06C4	3E06	STUP5:	LD	A, 6	; Reset motor line.
1117	06C6	3203E0		LD	(PORT3), A	
1118	06C9	3E07		LD	A, 7	; Set motor line.
1119	06CB	3203E0		LD	(PORT3), A	
1120	06CE	0D		DEC	C	
1121	06CF	C2B706		JP	NZ, STUP2	; Keep pulsing motor.
1122	06D2	CD0900		CALL	CRLF	
1123	06D5	7A		LD	A, D	
1124	06D6	FED7		CP	0D7H	; Punch on load?
1125	06D8	2808		JR	Z, STUP6	
1126	06DA	112207		LD	DE, M11	
1127	06DD	CD1800		CALL	STRING	

1128	06E0	180C		JR	STUP,	
1129	06E2	112907	STUP6:	LD	DE,M13	
1130	06E5	CD1800		CALL	STRING	
1131	06E8	112407		LD	DE,M12	
1132	06EB	CD1800		CALL	STRING	
1133	06EE	3A02E0	STUP7:	LD	A,(PORT2)	
1134	06F1	E610		AND	10H	; Look at sense line.
1135	06F3	C2BF06		JP	NZ,STUP3	; Motor on?
1136	06F6	CD440A		CALL	BREAK2	; Test for BREAK
1137	06F9	C2EE06		JP	NZ,STUP7	; No BREAK yet.
1138	06FC	37		SCF		
1139	06FD	C3C006		JP	STUP4	; BREAK abort.
1140						
1141	0700	F5	MOTOROFF:	PUSH	AF	
1142	0701	C5		PUSH	BC	
1143	0702	D5		PUSH	DE	
1144	0703	060A		LD	B,10	
1145	0705	3A02E0	MOT1:	LD	A,(PORT2)	
1146	0708	E610		AND	10H	; Sense.
1147	070A	2004		JR	NZ,MOT2	
1148	070C	D1		POP	DE	
1149	070D	C1		POP	BC	
1150	070E	F1		POP	AF	
1151	070F	C9		RET		; Motor's off.
1152	0710	3E06	MOT2:	LD	A,6	; Pulse motor.
1153	0712	3203E0		LD	(PORT3),A	
1154	0715	3E07		LD	A,7	
1155	0717	3203E0		LD	(PORT3),A	
1156	071A	05		DEC	B	
1157	071B	C20507		JP	NZ,MOT1	
1158	071E	B1		POP	DE	
1159	071F	C1		POP	BC	
1160	0720	F1		POP	AF	
1161	0721	C9		RET		
1162						
1163	0722	7F20	M11:	DB	7FH,' '	
1164	0724	504C4159	M12:	DB	'PLAY',CR	
1164	0728	0D				
1165	0729	7F205245	M13:	DB	7FH,' RECORD.',CR	
1165	072D	434F5244				
1165	0731	2E0D				
1166						
1167						
1168	0733	C5	COUNTBITS:	PUSH	BC	; Counts the
1169	0734	D5		PUSH	DE	; number of ONES in a
1170	0735	E5		PUSH	HL	; data block.
1171	0736	110000		LD	DE,0	; Zero count.
1172	0739	78	CBT2:	LD	A,B	
1173	073A	B1		OR	C	; Anymore?
1174	073B	200B		JR	NZ,CBT3	
1175	073D	EB		EX	DE,HL	
1176	073E	229711		LD	(BITCOUNT),HL	
1177	0741	229911		LD	(BITCOUNT2),HL	
1178	0744	E1		POP	HL	
1179	0745	D1		POP	DE	
1180	0746	C1		POP	BC	
1181	0747	C9		RET		
1182	0748	7E	CBT3:	LD	A,(HL)	; Pick up a byte.
1183	0749	E5		PUSH	HL	
1184	074A	2608		LD	H,8	; Bits per byte.

1185 074C 07	CBT4:	RLCA	; Shift into carry.
1186 074D 3001		JR NC,CBT5	
1187 074F 13		INC DE	; Bump count.
1188 0750 25	CBT5:	DEC H	
1189 0751 20F9		JR NZ,CBT4	
1190 0753 E1		POP HL	
1191 0754 23		INC HL	; Bump pointer.
1192 0755 0B		DEC BC	
1193 0756 C33907		JP CBT2	
1194			
1195 0759 3E0E	DELAY1:	LD A,14	
1196 075B 3D	DEL1:	DEC A	
1197 075C C25B07		JP NZ,DEL1	; Approx 125uS delay.
1198 075F C9		RET	
1199			
1200 0760 3E0D	DELAY2:	LD A,13	
1201 0762 3D	DEL2:	DEC A	
1202 0763 C26207		JP NZ,DEL2	; Approx 120uS.
1203 0766 C9		RET	
1204			
1205 0767 F5	SHORTBIT:	PUSH AF	
1206 0768 3E03		LD A,3	; Write line HI.
1207 076A 3203E0		LD (PORT3),A	
1208 076D CD5907		CALL DELAY1	
1209 0770 CD5907		CALL DELAY1	
1210 0773 3E02		LD A,2	; Write line LO.
1211 0775 3203E0		LD (PORT3),A	
1212 0778 CD5907		CALL DELAY1	
1213 077B CD5907		CALL DELAY1	
1214 077E F1		POP AF	
1215 077F C9		RET	
1216			
1217 0780 F5	LONGBIT:	PUSH AF	
1218 0781 3E03		LD A,3	
1219 0783 3203E0		LD (PORT3),A	
1220 0786 CD5907		CALL DELAY1	
1221 0789 CD5907		CALL DELAY1	
1222 078C CD5907		CALL DELAY1	
1223 078F CD5907		CALL DELAY1	
1224 0792 3E02		LD A,2	
1225 0794 3203E0		LD (PORT3),A	
1226 0797 CD5907		CALL DELAY1	
1227 079A CD5907		CALL DELAY1	
1228 079D CD5907		CALL DELAY1	
1229 07A0 CD6007		CALL DELAY2	
1230 07A3 F1		POP AF	
1231 07A4 C9		RET	
1232			
1233 07A5 C5	SRLOUT:	PUSH BC	
1234 07A6 0608		LD B,8	; Bit count.
1235 07A8 CD8007		CALL LONGBIT	; Start bit.
1236 07AB 07	SRLT2:	RLCA	; Shift bit to carry.
1237 07AC DC8007		CALL C,LONGBIT	; ONE = long.
1238 07AF D46707		CALL NC,SHORTBIT	; ZERO = short.
1239 07B2 05		DEC B	
1240 07B3 C2AB07		JP NZ,SRLT2	
1241 07B6 C1		POP BC	
1242 07B7 C9		RET	
1243			
1244 07B8 C5	LEADER:	PUSH BC	

1245	07B9	D5		PUSH	DE	
1246	07BA	78		LD	A,B	
1247	07EB	01F055		LD	BC,22000	; Leader lensth.
1248	07BE	112828		LD	DE,40*256+40	; Sync pulses.
1249	07C1	FECC		CP	OCCH	; Head on data?
1250	07C3	CACC07		JP	Z,LEAD2	
1251	07C6	01F82A		LD	BC,11000	
1252	07C9	111414		LD	DE,20*256+20	
1253	07CC	CD6707	LEAD2:	CALL	SHORTBIT	
1254	07CF	0B		DEC	BC	
1255	07D0	78		LD	A,B	
1256	07D1	B1		OR	C	
1257	07D2	20F8		JR	NZ,LEAD2	; Punch leader.
1258	07D4	CD8007	LEAD3:	CALL	LONGBIT	
1259	07D7	15		DEC	D	
1260	07D8	20FA		JR	NZ,LEAD3	; Punch sync.
1261	07DA	CD6707	LEAD4:	CALL	SHORTBIT	
1262	07DD	1D		DEC	E	
1263	07DE	20FA		JR	NZ,LEAD4	
1264	07E0	CD8007		CALL	LONGBIT	
1265	07E3	D1		POP	DE	
1266	07E4	C1		POP	BC	
1267	07E5	C9		RET		
1268						
1269	07E6	F5	User:	PUSH	AF	; Get a line of
1270	07E7	C5		PUSH	BC	; input from user.
1271	07E8	E5		PUSH	HL	
1272	07E9	D5		PUSH	DE	
1273	07EA	AF		XOR	A	
1274	07EB	329311		LD	(USERMODE),A	; Normal mode.
1275	07EE	CDB309	US2:	CALL	GETCHAR	; Get character.
1276	07F1	47		LD	B,A	; Save it.
1277	07F2	3A9D11		LD	A,(BEEPFLAG)	
1278	07F5	B7		OR	A	
1279	07F6	CCE502		CALL	Z,Beep	; Beep if on.
1280	07F9	78		LD	A,B	
1281	07FA	E6F0		AND	OFOH	; Strip low nibble.
1282	07FC	FEC0		CP	OC0H	; Control character?
1283	07FE	C2A608		JF	NZ,US15	; If not it's data.
1284	0801	78		LD	A,B	
1285	0802	FEC0		CP	OCDH	; Carriage Return?
1286	0804	CA5B08		JP	Z,US8	
1287	0807	FEC9		CP	OC9H	; Capitals?
1288	0809	CA2B08		JP	Z,US31	
1289	080C	FECA		CP	OCAH	; Small?
1290	080E	CA2B08		JP	Z,US31	
1291	0811	FECB		CP	OCBH	; BREAK?
1292	0813	CAB308		JP	Z,US16	
1293	0816	FEC8		CP	OC8H	; Insert?
1294	0818	CA3808		JP	Z,US5	
1295	081B	FEC7		CP	OC7H	; Delete?
1296	081D	CA2B08		JP	Z,US31	
1297	0820	3A9311		LD	A,(USERMODE)	
1298	0823	B7		OR	A	
1299	0824	FA4208		JP	M,US6	; Literal mode?
1300	0827	C23108		JP	NZ,US4	; Insert mode?
1301	082A	78	US3:	LD	A,B	
1302	082B	CDDC0D	US31:	CALL	CONTROL	; Else display.
1303	082E	C3EE07		JP	US2	; Keep going.
1304						

1305	0831	3D	US4:	DEC	A	
1306	0832	329311		LD	(USERMODE),A	; Remove insert.
1307	0835	C34208		JP	US6	
1308						
1309	0838	3A9311	US5:	LD	A,(USERMODE)	
1310	083B	3C		INC	A	; Set to insert mode.
1311	083C	329311		LD	(USERMODE),A	
1312	083F	C32A08		JP	US3	
1313						
1314	0842	78	US6:	LD	A,B	
1315	0843	CDA60D		CALL	SYNC	; Wait for Video blank.
1316	0846	CDB50D		CALL	DISPLAY	; Display on video.
1317	0849	FE62		CP	62H	; Was that a " ?
1318	084B	C2EE07		JP	NZ,US2	; If not,carry on.
1319	084E	3A9311		LD	A,(USERMODE)	
1320	0851	E680		AND	80H	; Isolate literal flag.
1321	0853	EE80		XOR	80H	; Flip to/from literal.
1322	0855	329311		LD	(USERMODE),A	
1323	0858	C3EE07		JP	US2	
1324						
1325	085B	2A7111	US8:	LD	HL,(CURSOR)	
1326	085E	5C		LD	E,H	
1327	085F	1600		LD	D,0	
1328	0861	217311		LD	HL,VIDMAP	
1329	0864	19		ADD	HL,DE	
1330	0865	EB		EX	DE,HL	
1331	0866	1A		LD	A,(DE)	; Look at flag for the
1332	0867	B7		OR	A	; line to see if line is
1333	0868	012800		LD	BC,40	; longer than 40 chrs.
1334	086B	2A7111		LD	HL,(CURSOR)	
1335	086E	C27A08		JP	NZ,US9	; Yes overflowed.
1336	0871	13		INC	DE	
1337	0872	1A		LD	A,(DE)	
1338	0873	B7		OR	A	
1339	0874	CA7D08		JP	Z,US11	
1340	0877	C37B08		JP	US10	
1341	087A	25	US9:	DEC	H	; Back up a line.
1342	087B	0E50	US10:	LD	C,80	; Two lines to do.
1343	087D	2E00	US11:	LD	L,0	; Start of line.
1344	087F	CDB40F		CALL	FDS1	; Find position.
1345	0882	D1		POP	DE	
1346	0883	D5		PUSH	DE	; Pick up textbuffer.
1347	0884	C5		PUSH	BC	; Save length
1348	0885	CDA60D		CALL	SYNC	
1349	0888	EDB0		LDIR		; Move from video to textbuff
1350	088A	C1		POP	BC	
1351	088B	E1		POP	HL	
1352	088C	E5		PUSH	HL	; Pick up textbuffer.
1353	088D	41		LD	B,C	; Length.
1354	088E	7E	US12:	LD	A,(HL)	; Pick up char.
1355	088F	CDCE0B		CALL	NORMAL	; Turn to ASCII.
1356	0892	77		LD	(HL),A	; Put it back
1357	0893	23		INC	HL	
1358	0894	10F8		DJNZ	US12	; Correct whole line.
1359	0896	360D	US13:	LD	(HL),CR	; Terminate
1360	0898	2B		DEC	HL	
1361	0899	7E		LD	A,(HL)	
1362	089A	FE20		CP	BLANK	
1363	089C	28F8		JR	Z,US13	; Scrub trailing blanks.
1364	089E	CD0600	US14:	CALL	CRLF2	

1365	08A1	D1		POP	DE	
1366	08A2	E1		POP	HL	
1367	08A3	C1		POP	BC	
1368	08A4	F1		POP	AF	
1369	08A5	C9		RET		
1370						
1371	08A6	3A9311	US15:	LD	A, (USERMODE)	
1372	08A9	B7		OR	A	
1373	08AA	FA4208		JP	M, US6	; Literal?
1374	08AD	CA4208		JP	Z, US6	; Normal?
1375	08B0	C33108		JP	US4	; Insert
1376						
1377	08B3	E1	US16:	POP	HL	
1378	08B4	E5		PUSH	HL	; Pick up textbuffer
1379	08B5	361B		LD	(HL), ESCAPE	; Crash out
1380	08B7	23		INC	HL	
1381	08B8	360D		LD	(HL), CR	
1382	08BA	C39E08		JP	US14	
1383						
1384	08BD	CDCA08	Keyboard:	CALL	WHICHKEY	
1385	08C0	FEF0		CP	OFOH	; Keypressed?
1386	08C2	2002		JR	NZ, KBD2	
1387	08C4	AF		XOR	A	
1388	08C5	C9		RET		
1389	08C6	CDCE08	KBD2:	CALL	NORMAL	; Turn to ASCII
1390	08C9	C9		RET		
1391						
1392	08CA	C5	WHICHKEY:	PUSH	BC	; Find which key
1393	08CB	D5		PUSH	DE	; has been pressed.
1394	08CC	E5		PUSH	HL	
1395	08CD	CD500A		CALL	SCAN	
1396	08D0	78		LD	A, B	
1397	08D1	07		RLCA		; Shift keypress flag.
1398	08D2	3806		JR	C, WH2	
1399	08D4	3EF0		LD	A, OFOH	; No keys
1400	08D6	E1	WH1:	POP	HL	
1401	08D7	D1		POP	DE	
1402	08D8	C1		POP	BC	
1403	08D9	C9		RET		
1404						
1405	08DA	07	WH2:	RLCA		; Test for SHIFT key.
1406	08DB	D2EC08		JP	NC, WH3	
1407	08DE	0600		LD	B, 0	
1408	08E0	210800		LD	HL, 8	; Add shift offset
1409	08E3	09		ADD	HL, BC	; Form pointer
1410	08E4	11C90A	WH21:	LD	DE, CAPTABLE	; Capitals
1411	08E7	19		ADD	HL, DE	; Point to entry
1412	08E8	7E		LD	A, (HL)	; Pick up code
1413	08E9	C3D608		JP	WH1	
1414						
1415	08EC	3A7011	WH3:	LD	A, (CASEFLAG)	; Check case
1416	08EF	B7		OR	A	
1417	08F0	C2FD08		JP	NZ, WH4	; NZ = Small
1418	08F3	0600		LD	B, 0	
1419	08F5	21C90A		LD	HL, CAPTABLE	
1420	08F8	09		ADD	HL, BC	; Point entry
1421	08F9	7E		LD	A, (HL)	
1422	08FA	C3D608		JP	WH1	
1423						
1424	08FD	79	WH4:	LD	A, C	; Pick up index

1425	08FE	E6F0		AND	OFOH	; Strip column
1426	0900	0F		RRCA		; Pack together
1427	0901	47		LD	B,A	
1428	0902	79		LD	A,C	; Pick up index
1429	0903	E60F		AND	OFOH	; Strip row
1430	0905	80		ADD	A,B	; Form packed index
1431	0906	C6A0		ADD	A,0A0H	; Point small table
1432	0908	6F		LD	L,A	
1433	0909	2600		LD	H,0	
1434	090B	C3E408		JP	WH21	; Go pick up character
1435						
1436	090E	AF	Crlf2:	XOR	A	
1437	090F	329411		LD	(CHARCOUNT),A	
1438	0912	3ECD		LD	A,0CDH	; Newline control char.
1439	0914	CDDCOD		CALL	CONTROL	
1440	0917	C9		RET		
1441						
1442	0918	3A9411	Crlf:	LD	A,(CHARCOUNT)	
1443	091B	B7		OR	A	
1444	091C	C8		RET	Z	
1445	091D	C30600		JP	CRLF2	
1446						
1447	0920	3E20	Space:	LD	A,BLANK	
1448	0922	CD3509		CALL	Video	
1449	0925	C9		RET		
1450						
1451	0926	CD0C00	Tab:	CALL	SPACE	
1452	0929	3A9411		LD	A,(CHARCOUNT)	
1453	092C	B7		OR	A	
1454	092D	C8		RET	Z	
1455	092E	D60A	TAB2:	SUB	10	
1456	0930	38F4		JR	C,Tab	
1457	0932	20FA		JR	NZ,TAB2	
1458	0934	C9		RET		
1459						
1460	0935	FE0D	Video:	CP	CR	
1461	0937	CA0E09		JP	Z,Crlf2	
1462	093A	C5		PUSH	BC	
1463	093B	4F		LD	C,A	
1464	093C	47		LD	B,A	
1465	093D	CDA60D		CALL	SYNC	; Wait for blanking
1466	0940	CD4609		CALL	DECODE	
1467	0943	78		LD	A,B	
1468	0944	C1		POP	BC	
1469	0945	C9		RET		
1470						
1471	0946	79	DECODE:	LD	A,C	
1472	0947	CDB90B		CALL	SPECIAL	; Turn ASCII to
1473	094A	4F		LD	C,A	; special character.
1474	094B	E6F0		AND	OFOH	
1475	094D	FEF0		CP	OFOH	; Unused character?
1476	094F	C8		RET	Z	
1477	0950	FEC0		CP	OC0H	; Control character?
1478	0952	79		LD	A,C	
1479	0953	C27009		JP	NZ,DATACHAR	; Else data
1480	0956	FEC7		CP	OC7H	
1481	0958	D27009		JP	NC,DATACHAR	
1482	095B	CDDCOD		CALL	CONTROL	
1483	095E	FEC3		CP	OC3H	; Cursor right?
1484	0960	CA7309		JP	Z,DCHAR2	

1485	0963	FEC5		CP	0C5H	; Home?
1486	0965	CA6B09		JP	Z,DEC2	
1487	0968	FEC6		CP	0C6H	; Clear?
1488	096A	C0		RET	NZ	
1489	096B	AF	DEC2:	XOR	A	
1490	096C	329411		LD	(CHARCOUNT),A	
1491	096F	C9		RET		
1492						
1493	0970	CDB50D	DATACHAR:	CALL	DISPLAY	; Display it
1494	0973	3A9411	DCHAR2:	LD	A,(CHARCOUNT)	
1495	0976	3C		INC	A	
1496	0977	FE50		CP	80	
1497	0979	3802		JR	C,DCHAR3	
1498	097B	D650		SUB	80	
1499	097D	329411	DCHAR3:	LD	(CHARCOUNT),A	
1500	0980	C9		RET		
1501						
1502	0981	F5	Message:	PUSH	AF	
1503	0982	C5		PUSH	BC	
1504	0983	D5		PUSH	DE	
1505	0984	0605	MS1:	LD	B,5	
1506	0986	CDA60D		CALL	SYNC	; Wait for blanking
1507	0989	1A	MS2:	LD	A,(DE)	
1508	098A	FE0D		CP	CR	; Terminator?
1509	098C	CADFOF		JP	Z,POP2	
1510	098F	4F		LD	C,A	
1511	0990	CD4609		CALL	DECODE	; Display character
1512	0993	13		INC	DE	
1513	0994	10F3		DJNZ	MS2	; Time for more?
1514	0996	C38409		JP	MS1	
1515						
1516	0999	F5	Strings:	PUSH	AF	
1517	099A	C5		PUSH	BC	
1518	099B	D5		PUSH	DE	
1519	099C	0605	STR1:	LD	B,5	
1520	099E	CDA60D		CALL	SYNC	
1521	09A1	1A	STR2:	LD	A,(DE)	
1522	09A2	FE0D		CP	CR	
1523	09A4	CADFOF		JP	Z,POP2	
1524	09A7	CDB90B		CALL	SPECIAL	; Turn to special
1525	09AA	CD7009		CALL	DATACHAR	; Display all chars
1526	09AD	13		INC	DE	
1527	09AE	10F1		DJNZ	STR2	
1528	09B0	C39C09		JP	STR1	
1529						
1530	09B3	C5	GETCHAR:	PUSH	BC	
1531	09B4	D5		PUSH	DE	
1532	09B5	E5		PUSH	HL	
1533	09B6	CDB10F		CALL	POSITION	; Video position
1534	09B9	CDA60D		CALL	SYNC	
1535	09BC	7E		LD	A,(HL)	; Pick up video char
1536	09BD	328E11		LD	(VIDCHAR),A	; Save it
1537	09C0	228F11		LD	(VIDPOS),HL	; Save position
1538	09C3	219211		LD	HL,CURCHAR	; Char at cursor
1539	09C6	36EF		LD	(HL),0EFH	; Cursor there
1540	09C8	AF		XOR	A	
1541	09C9	3200E0		LD	(PORT0),A	; Reset blink timer
1542	09CC	329111		LD	(ONOFF),A	
1543	09CF	2F		CPL		
1544	09D0	3200E0		LD	(PORT0),A	

1545	09D3	1614	CHR2:	LD	D, 20	
1546	09D5	CDFF09	CHR3:	CALL	BLINK	; Blink cursor
1547	09D8	CD500A		CALL	SCAN	; Scan keyboard
1548	09DB	78		LD	A, B	
1549	09DC	07		RLCA		
1550	09DD	DAD309		JP	C, CHR2	; Wait for bounce
1551	09E0	15		DEC	D	
1552	09E1	C2D509		JP	NZ, CHR3	; End of bounce
1553	09E4	CDFF09	CHR4:	CALL	BLINK	
1554	09E7	CDCA08		CALL	WHICHKEY	
1555	09EA	FEF0		CF	OFOH	; Good key?
1556	09EC	CAE409		JP	Z, CHR4	
1557	09EF	F5		PUSH	AF	
1558	09F0	CDA60D		CALL	SYNC	
1559	09F3	3A8E11		LD	A, (VIDCHAR)	
1560	09F6	2A8F11		LD	HL, (VIDPOS)	
1561	09F9	77		LD	(HL), A	; Restore character
1562	09FA	F1		POP	AF	
1563	09FB	E1		POP	HL	
1564	09FC	D1		POP	DE	
1565	09FD	C1		POP	BC	
1566	09FE	C9		RET		
1567						
1568	09FF	F5	BLINK:	PUSH	AF	
1569	0A00	E5		PUSH	HL	
1570	0A01	3A02E0		LD	A, (PORT2)	; Look at blink timer
1571	0A04	07		RLCA		
1572	0A05	07		RLCA		; Shift blink bit
1573	0A06	DA250A		JP	C, BLINK3	
1574	0A09	3A9111		LD	A, (ONOFF)	
1575	0A0C	0F		RRCA		; Cursor on or off?
1576	0A0D	DA220A		JP	C, BLINK2	
1577	0A10	3A9211		LD	A, (CURCHAR)	
1578	0A13	2A8F11	BLINK1:	LD	HL, (VIDPOS)	
1579	0A16	CDA60D		CALL	SYNC	
1580	0A19	77		LD	(HL), A	
1581	0A1A	3A9111		LD	A, (ONOFF)	
1582	0A1D	EE01		XOR	1	; Flip on off flag
1583	0A1F	329111		LD	(ONOFF), A	
1584	0A22	E1	BLINK2:	POP	HL	
1585	0A23	F1		POP	AF	
1586	0A24	C9		RET		
1587	0A25	3A9111	BLINK3:	LD	A, (ONOFF)	
1588	0A28	0F		RRCA		
1589	0A29	D2220A		JP	NC, BLINK2	
1590	0A2C	3A8E11		LD	A, (VIDCHAR)	
1591	0A2F	C3130A		JP	BLINK1	; Replace character
1592						
1593	0A32	3EF8	Break:	LD	A, 0F8H	
1594	0A34	3200E0		LD	(PORT0), A	; Activate SHIFT row
1595	0A37	00		NOP		
1596	0A38	3A01E0		LD	A, (PORT1)	
1597	0A3B	2F		CPL		
1598	0A3C	E621		AND	21H	; Either SHIFT pressed?
1599	0A3E	C2440A		JP	NZ, BREAK2	
1600	0A41	C601		ADD	A, 1	
1601	0A43	C9		RET		
1602	0A44	3EF9	BREAK2:	LD	A, 0F9H	
1603	0A46	3200E0		LD	(PORT0), A	; BREAK row
1604	0A49	00		NOP		

1605	0A4A	3A01E0	LD	A, (PORT1)	
1606	0A4D	E608	AND	8	; Isolate BREAK
1607	0A4F	C9	RET		
1608					
1609	0A50	D5	SCAN:	PUSH DE	
1610	0A51	E5		PUSH HL	
1611	0A52	06FA		LD B, 0FAH	; Row counter to 10
1612	0A54	1600		LD D, 0	; Special flags
1613	0A56	05	SCAN2:	DEC B	; Next row
1614	0A57	78		LD A, B	
1615	0A58	3200E0		LD (PORT0), A	; Output row
1616	0A5B	FEEF		CP 0EFH	; Finished?
1617	0A5D	C2640A		JP NZ, SCAN3	
1618	0A60	42		LD B, D	; Flags
1619	0A61	D1		POP DE	
1620	0A62	E1		POP HL	
1621	0A63	C9		RET	
1622					
1623	0A64	FEF8	SCAN3:	CP 0F8H	; SHIFT row?
1624	0A66	CAB20A		JP Z, SCAN12	
1625	0A69	3A01E0		LD A, (PORT1)	; Read columns
1626	0A6C	2F		CPL	; Make active HI
1627	0A6D	B7		OR A	; Any keys?
1628	0A6E	CA560A		JP Z, SCAN2	; Try next
1629	0A71	5F	SCAN31:	LD E, A	; Save columns
1630	0A72	7A		LD A, D	
1631	0A73	F680		OR 80H	; Key press flag
1632	0A75	57		LD D, A	
1633	0A76	21AD0A		LD HL, SCAN11	; Forward jump
1634	0A79	78		LD A, B	; Row count
1635	0A7A	E60F		AND 0FH	; Strip garbage
1636	0A7C	07		RLCA	; Shift
1637	0A7D	07		RLCA	
1638	0A7E	07		RLCA	
1639	0A7F	07		RLCA	
1640	0A80	4F		LD C, A	; C = Row
1641	0A81	7B		LD A, E	; Change column to integer
1642	0A82	07		RLCA	; Column 7?
1643	0A83	3003		JR NC, SCAN4	
1644	0A85	3E07		LD A, 7	
1645	0A87	E9		JP (HL)	
1646	0A88	07	SCAN4:	RLCA	; Column 6?
1647	0A89	3003		JR NC, SCAN5	
1648	0A8B	3E06		LD A, 6	
1649	0A8D	E9		JP (HL)	
1650	0A8E	07	SCAN5:	RLCA	; Column 5?
1651	0A8F	3003		JR NC, SCAN6	
1652	0A91	3E05		LD A, 5	
1653	0A93	E9		JP (HL)	
1654	0A94	07	SCAN6:	RLCA	; Column 4?
1655	0A95	3003		JR NC, SCAN7	
1656	0A97	3E04		LD A, 4	
1657	0A99	E9		JP (HL)	
1658	0A9A	07	SCAN7:	RLCA	; Column 3?
1659	0A9B	3003		JR NC, SCAN8	
1660	0A9D	3E03		LD A, 3	
1661	0A9F	E9		JP (HL)	
1662	0AA0	07	SCAN8:	RLCA	; Column 2?
1663	0AA1	3003		JR NC, SCAN9	
1664	0AA3	3E02		LD A, 2	

1665	OAA5	E9		JP	(HL)	
1666	OAA6	07	SCAN9:	RLCA		; Column 1?
1667	OAA7	3003		JR	NC,SCAN10	
1668	OAA9	3E01		LD	A,1	
1669	OAA8	E9		JP	(HL)	
1670	OAAC	AF	SCAN10:	XOR	A	; Column 0
1671	OAAD	81	SCAN11:	ADD	A,C	; Add to row
1672	OAAE	4F		LD	C,A	; C = Index
1673	OAAF	C3560A		JP	SCAN2	
1674						
1675	OAB2	3A01E0	SCAN12:	LD	A,(PORT1)	; Columns
1676	OAB5	2F		CPL		
1677	OAB6	5F		LD	E,A	
1678	OAB7	E621		AND	21H	; Isolate SHIFT keys
1679	OAB9	CAC00A		JP	Z,SCAN13	; No SHIFT
1680	OABC	7A		LD	A,D	
1681	OABD	F640		OR	40H	; Set SHIFT flag
1682	OABF	57		LD	D,A	
1683	OACO	7B	SCAN13:	LD	A,E	
1684	OAC1	E6DE		AND	0DEH	; Anything else?
1685	OAC3	CA560A		JP	Z,SCAN2	
1686	OAC6	C3710A		JP	SCAN31	; Carry on
1687						
1688	OAC9	21232527	CAPTABLE:	DB	21H,23H,25H,27H	; row 0
1689	OACD	292A1D1F		DB	29H,2AH,1DH,1FH	
1690						
1691	OAD1	61636567		DB	61H,63H,65H,67H	; SHIFT row 0
1692	OAD5	696A5D5F		DB	69H,6AH,5DH,5FH	
1693						
1694	OAD9	22242628		DB	22H,24H,26H,28H	; row 1
1695	OADD	201C1ED1		DB	20H,1CH,1EH,0D1H	
1696						
1697	OAE1	62646668		DB	62H,64H,66H,68H	; SHIFT row 1
1698	OAE5	605C5ED0		DB	60H,5CH,5EH,0D0H	
1699						
1700	OAE9	11051415		DB	11H,05H,14H,15H	; row 2
1701	OAED	0F2B3133		DB	0FH,2BH,31H,33H	
1702						
1703	OAF1	51455455		DB	51H,45H,54H,55H	; SHIFT row 2
1704	OAF5	4F6B7173		DB	4FH,6BH,71H,73H	
1705						
1706	OAF9	17121909		DB	17H,12H,19H,09H	; row 3
1707	OAFD	103032D3		DB	10H,30H,32H,0D3H	
1708						
1709	OB01	57525949		DB	57H,52H,59H,49H	; SHIFT row 3
1710	OB05	507072D2		DB	50H,70H,72H,0D2H	
1711						
1712	OB09	0104070A		DB	01H,04H,07H,0AH	; row 4
1713	OB0D	0C1B3537		DB	0CH,1BH,35H,37H	
1714						
1715	OB11	4144474A		DB	41H,44H,47H,4AH	; SHIFT row 4
1716	OB15	4C5B7577		DB	4CH,5BH,75H,77H	
1717						
1718	OB19	1306080B		DB	13H,06H,08H,0BH	; row 5
1719	OB1D	2C3436D5		DB	2CH,34H,36H,0D5H	
1720						
1721	OB21	5346484B		DB	53H,46H,48H,4BH	; SHIFT row 5
1722	OB25	6C7476D4		DB	6CH,74H,76H,0D4H	
1723						
1724	OB29	1A03020D		DB	1AH,03H,02H,0DH	; row 6

1725	0B2D	2EC9393B	DB	2EH, 0C9H, 39H, 3BH
1726				
1727	0B31	5A43424D	DB	5AH, 43H, 42H, 4DH ; SHIFT row 6
1728	0B35	6ECA797B	DB	6EH, 0CAH, 79H, 7BH
1729				
1730	0B39	18160E2F	DB	18H, 16H, 0EH, 2FH ; row 7
1731	0B3D	2D383AD7	DB	2DH, 38H, 3AH, 0D7H
1732				
1733	0B41	58564E6F	DB	58H, 56H, 4EH, 6FH ; SHIFT row 7
1734	0B45	6D787AD6	DB	6DH, 78H, 7AH, 0D6H
1735				
1736	0B49	FOC7FOC3	DB	0FOH, 0C7H, 0FOH, 0C3H; row 8
1737	0B4D	CDF03D3F	DB	0CDH, 0FOH, 3DH, 3FH
1738				
1739	0B51	FOC8FOC4	DB	0FOH, 0C8H, 0FOH, 0C4H; SHIFT row 8
1740	0B55	CDF07D7F	DB	0CDH, 0FOH, 7DH, 7FH
1741				
1742	0B59	C500C1FO	DB	0C5H, 00H, 0C1H, 0FOH; row 9
1743	0B5D	F03C3EDC	DB	0FOH, 3CH, 3EH, 0DCH
1744				
1745	0B61	C600C2CB	DB	0C6H, 00H, 0C2H, 0CBH; SHIFT row 9
1746	0B65	F07C7ED8	DB	0FOH, 7CH, 7EH, 0D8H
1747				
1748				
1749				
1750				
1751	0B69	A1A3A5A7	DB	0A1H, 0A3H, 0A5H, 0A7H; row 0
1752	0B6D	A9AA9D9F	DB	0A9H, 0AAH, 9DH, 9FH
1753				
1754	0B71	A2A4A6A8	DB	0A2H, 0A4H, 0A6H, 0A8H; row 1
1755	0B75	A09C9EDD	DB	0A0H, 9CH, 9EH, 0DDH
1756				
1757	0B79	91859495	DB	91H, 85H, 94H, 95H ; row 2
1758	0B7D	8FA5B1B3	DB	8FH, 0ABH, 0B1H, 0B3H
1759				
1760	0B81	97929989	DB	97H, 92H, 99H, 89H ; row 3
1761	0B85	90B0B2DE	DB	90H, 0B0H, 0B2H, 0DEH
1762				
1763	0B89	8184878A	DB	81H, 84H, 87H, 8AH, ; row 4
1764	0B8D	8C9BB5B7	DB	8CH, 9BH, 0B5H, 0B7H
1765				
1766	0B91	9386888B	DB	93H, 86H, 88H, 8BH ; row 5
1767	0B95	ACB4B6D9	DB	0ACH, 0B4H, 0B6H, 0D9H
1768				
1769	0B99	9A83828D	DB	9AH, 83H, 82H, 8DH ; row 6
1770	0B9D	AEC9B9BB	DB	0AEH, 0C9H, 0B9H, 0BBH
1771				
1772	0BA1	98968EAF	DB	98H, 96H, 8EH, 0AFH; row 7
1773	0BA5	ADB8BADA	DB	0ADH, 0B8H, 0BAH, 0DAH
1774				
1775	0BA9	FOC7FOC3	DB	0FOH, 0C7H, 0FOH, 0C3H; row 8
1776	0BAD	CDF08DBF	DB	0CDH, 0FOH, 0BDH, 0BFH
1777				
1778	0BB1	C500C1FO	DB	0C5H, 00H, 0C1H, 0FOH; row 9
1779	0BB5	F0BCBEDB	DB	0FOH, 0BCH, 0BEH, 0DBH
1780				
1781	0BB9	D610	SPECIAL: SUB	16
1782	0BBB	D2C10B	JP	NC, SPEC2
1783	0BBE	3EF0	LD	A, 0FOH ; Ignore ASCII <16
1784	0BC0	C9	RET	

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1785
1786 OBC1 C5      SPEC2:  PUSH BC
1787 OBC2 E5      PUSH HL
1788 OBC3 21D60B  LD      HL,SPECTABLE      ; Special chars
1789 OBC6 4F      SPEC3:  LD      C,A
1790 OBC7 0600    LD      B,0
1791 OBC9 09      ADD     HL,BC              ; Point to entry
1792 OBCE 7E      LD      A,(HL)           ; Pick it up
1793 OCB8 E1      POP     HL
1794 OBCC C1      POP     BC
1795 OBCE C9      RET
1796
1797 OBCE C5      NORMAL:  PUSH BC
1798 OBCF E5      PUSH HL
1799 OBDO 21C60C  LD      HL,NORMTABLE      ; Normal ASCII
1800 OBD3 C3C60B  JP      SPEC3              ; Do the rest
1801
1802
1803      ; Table of special characters used
1804      ; to drive the non-standard
1805      ; character generator in the video
1806      ; display.ASCII characters less
1807      ; than decimal 16 are ignored so
1808      ; the first entry in the table is
1809      ; the equivalent of ASCII 16.
1810      ; For example ASCII 32 (Space) is
1811      ; converted to 00H.
1812
1813 OBD6 F0C1C2C3 SPECTABLE: DB      0F0H,0C1H,0C2H,0C3H
1814 OBDA C4C5C6F0 DB      0C4H,0C5H,0C6H,0F0H
1815 OBDE F0F0F0F0 DB      0F0H,0F0H,0F0H,0F0H
1816 OBE2 F0F0F0F0 DB      0F0H,0F0H,0F0H,0F0H
1817 OBE6 00616263 DB      00H,61H,62H,63H
1818 OBEA 64656667 DB      64H,65H,66H,67H
1819 OBEF 68696B6A DB      68H,69H,6BH,6AH
1820 OBF2 2F2A2E2D DB      2FH,2AH,2EH,2DH
1821 OBF6 20212223 DB      20H,21H,22H,23H
1822 OBFA 24252627 DB      24H,25H,26H,27H
1823 OBFE 28294F2C DB      28H,29H,4FH,2CH
1824 OC02 512B5749 DB      51H,2BH,57H,49H
1825 OC06 55010203 DB      55H,01H,02H,03H
1826 OC0A 04050607 DB      04H,05H,06H,07H
1827 OC0E 08090A0B DB      08H,09H,0AH,0BH
1828 OC12 0C0D0E0F DB      0CH,0DH,0EH,0FH
1829 OC16 10111213 DB      10H,11H,12H,13H
1830 OC1A 14151617 DB      14H,15H,16H,17H
1831 OC1E 18191A52 DB      18H,19H,1AH,52H
1832 OC22 59545045 DB      59H,54H,50H,45H
1833 OC26 C7C8C9CA DB      0C7H,0C8H,0C9H,0CAH
1834 OC2A CBCCCDCE DB      0CBH,0CCH,0CDH,0CEH
1835 OC2E CFDFE7E8 DB      0CFH,0DFH,0E7H,0E8H
1836 OC32 E9EAECEB DB      0E9H,0EAH,0ECH,0EDH
1837 OC36 D0D1D2D3 DB      0D0H,0D1H,0D2H,0D3H
1838 OC3A D4D5D6D7 DB      0D4H,0D5H,0D6H,0D7H
1839 OC3E D8D9DADB DB      0D8H,0D9H,0DAH,0DBH
1840 OC42 DCDDDECO DB      0DCH,0DDH,0DEH,0COH
1841 OC46 00BD9DB1 DB      00H,0BDH,9DH,0B1H
1842 OC4A B5B9B49E DB      0B5H,0B9H,0B4H,9EH
1843 OC4E B2F6BABE DB      0B2H,0B6H,0BAH,0BEH
1844 OC52 9FB3B7BB DB      9FH,0B3H,0B7H,0BBH

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1845	OC56	BFA385A4	DB	0BFH, 0A3H, 85H, 0A4H
1846	OC5A	A5A69487	DB	0A5H, 0A6H, 94H, 87H
1847	OC5E	889C8298	DB	88H, 9CH, 82H, 98H
1848	OC62	849290B3	DB	84H, 92H, 90H, 83H
1849	OC66	91819A97	DB	91H, 81H, 9AH, 97H
1850	OC6A	939589A1	DB	93H, 95H, 89H, 0A1H
1851	OC6E	AF8B8696	DB	0AFH, 8BH, 86H, 96H
1852	OC72	A2ABAA8A	DB	0A2H, 0ABH, 0AAH, 8AH
1853	OC76	8EB0AD8D	DB	8EH, 0B0H, 0ADH, 8DH
1854	OC7A	A7A8A98F	DB	0A7H, 0A8H, 0A9H, 8FH
1855	OC7E	8CAEAC9B	DB	8CH, 0AEH, 0ACH, 9BH
1856	OC82	A099BCB8	DB	0A0H, 99H, 0BCH, 0B8H
1857	OC86	003B3A70	DB	00H, 3BH, 3AH, 70H
1858	OC8A	3C715A3D	DB	3CH, 71H, 5AH, 3DH
1859	OC8E	43563F1E	DB	43H, 56H, 3FH, 1EH
1860	OC92	4A1C5D3E	DB	4AH, 1CH, 5DH, 3EH
1861	OC96	5C1F5F5E	DB	5CH, 1FH, 5FH, 5EH
1862	OC9A	377B7F36	DB	37H, 7BH, 7FH, 36H
1863	OC9E	7A7E334B	DB	7AH, 7EH, 33H, 4BH
1864	OCA2	4C1D6C5B	DB	4CH, 1DH, 6CH, 5BH
1865	OCA6	78413534	DB	78H, 41H, 35H, 34H
1866	OCAA	74303875	DB	74H, 30H, 38H, 75H
1867	OCAE	394D6F6E	DB	39H, 4DH, 6FH, 6EH
1868	OCB2	32777672	DB	32H, 77H, 76H, 72H
1869	OCB6	73477C53	DB	73H, 47H, 7CH, 53H
1870	OCBA	314E6D48	DB	31H, 4EH, 6DH, 48H
1871	OCBE	467D441B	DB	46H, 7DH, 44H, 1BH
1872	OCC2	58794260	DB	58H, 79H, 42H, 60H

1873

1874

; Table to convert the special

1875

; characters to 'normal' ASCII.

1876

; Table starts at zero which

1877

; produces ASCII '32 (Space),

1878

1879	OCC6	20414243	NORMTABLE: DB	' ABC'
1880	OCCA	44454647	DB	' DEFG'
1881	OCCE	48494A4B	DB	' HIJK'
1882	OCD2	4C4D4E4F	DB	' LMNO'
1883	OCD6	50515253	DB	' PQRS'
1884	OCDA	54555657	DB	' TUVW'
1885	OCDE	58595AFB	DB	' XYZ', OFBH
1886	OCE2	CDDDCBD1	DB	0CDH, 0DDH, 0CEH, 0D1H
1887	OCE6	30313233	DB	' 0123'
1888	OCEA	34353637	DB	' 4567'
1889	OCEE	38392D3D	DB	' 89-='
1890	OCF2	3B2F2E2C	DB	3BH, 2FH, 2EH, 2CH
1891	OCF6	E5F4ECDA	DB	0E5H, 0F4H, 0ECH, 0DAH
1892	OCFA	E3E2D7D4	DB	0E3H, 0E2H, 0D7H, 0D4H
1893	OCFE	E6E8C2C1	DB	0E6H, 0E8H, 0C2H, 0C1H
1894	ODO2	C4C7CFCA	DB	0C4H, 0C7H, 0CFH, 0CAH
1895	ODO6	20E1FEC8	DB	20H, 0E1H, 0FEH, 0C8H
1896	ODOA	FA5FF8F1	DB	0FAH, 5FH, 0F8H, 0F1H
1897	ODOE	F73FCCDB	DB	0F7H, 3FH, 0CCH, 0DBH
1898	OD12	DCE9F53A	DB	0DCH, 0E9H, 0F5H, 3AH
1899	OD16	5E3C5BF3	DB	5EH, 3CH, 5BH, 0F3H
1900	OD1A	5D40C93E	DB	5DH, 40H, 0C9H, 3EH
1901	OD1E	FC5CC6DF	DB	0FCH, 5CH, 0C6H, 0DFH
1902	OD22	D0CED3D2	DB	0D0H, 0CEH, 0D3H, 0D2H
1903	OD26	FF212223	DB	0FFH, 21H, 22H, 23H
1904	OD2A	24252627	DB	24H, 25H, 26H, 27H

1905	0D2E	28292B2A		DB	28H, 29H, 2EH, 2AH
1906	0D32	DEF6EBEA		DB	0DEH, 0F6H, 0EBH, 0EAH
1907	0D36	C3C5EFF0		DB	0C3H, 0C5H, 0EFH, 0F0H
1908	0D3A	E4E7EEED		DB	0E4H, 0E7H, 0EEH, 0EDH
1909	0D3E	E0FDD8D5		DB	0E0H, 0FDH, 0D8H, 0D5H
1910	0D42	F2F9D9D6		DB	0F2H, 0F9H, 0D9H, 0D6H
1911	0D46	20A19A9F		DB	20H, 0A1H, 9AH, 9FH
1912	0D4A	9C92AA97		DB	9CH, 92H, 0AAH, 97H
1913	0D4E	98A6AFA9		DB	98H, 0A6H, 0AFH, 0A9H
1914	0D52	B8B3B0B7		DB	0B8H, 0B3H, 0B0H, 0B7H
1915	0D56	9EA09DA4		DB	9EH, 0A0H, 9DH, 0A4H
1916	0D5A	96A5ABA3		DB	96H, 0A5H, 0ABH, 0A3H
1917	0D5E	9BBDAA2BB		DB	9BH, 0BDH, 0A2H, 0BBH
1918	0D62	9982878C		DB	99H, 82H, 87H, 8CH
1919	0D66	BCA7AC91		DB	0BCH, 0A7H, 0ACH, 91H
1920	0D6A	939495B4		DB	93H, 94H, 95H, 0B4H
1921	0D6E	B5B6AEAD		DB	0B5H, 0B6H, 0AEH, 0ADH
1922	0D72	BAB2B9A8		DB	0BAH, 0B2H, 0B9H, 0A8H
1923	0D76	B183888D		DB	0B1H, 83H, 88H, 8DH
1924	0D7A	8684898E		DB	86H, 84H, 89H, 8EH
1925	0D7E	BF858A8F		DB	0BFH, 85H, 8AH, 8FH
1926	0D82	BE818B90		DB	0BEH, 81H, 8BH, 90H
1927	0D86	7F111213		DB	7FH, 11H, 12H, 13H
1928	0D8A	14151660		DB	14H, 15H, 16H, 60H
1929	0D8E	61626364		DB	61H, 62H, 63H, 64H
1930	0D92	65666768		DB	65H, 66H, 67H, 68H
1931	0D96	70717273		DB	70H, 71H, 72H, 73H
1932	0D9A	74757677		DB	74H, 75H, 76H, 77H
1933	0D9E	78797A7B		DB	78H, 79H, 7AH, 7BH
1934	0DA2	7C7D7E69		DB	7CH, 7DH, 7EH, 69H
1935					
1936	0DA6	F5	SYNC:	PUSH	AF
1937	0DA7	3A02E0	SYN2:	LD	A, (PORT2)
1938	0DAA	07		RLCA	; Look at video blanking
1939	0DAB	30FA		JR	NC, SYN2
1940	0DAD	3A02E0	SYN3:	LD	A, (PORT2)
1941	0DB0	07		RLCA	
1942	0DB1	38FA		JR	C, SYN3 ; Wait for blanking pulse
1943	0DB3	F1		POP	AF
1944	0DB4	C9		RET	
1945					
1946	0DB5	F5	DISPLAY:	PUSH	AF
1947	0DB6	C5		PUSH	BC
1948	0DB7	D5		PUSH	DE
1949	0DB8	E5		PUSH	HL
1950	0DB9	47		LD	B, A
1951	0DBA	CDB10F		CALL	POSITION
1952	0DBD	70		LD	(HL), B ; Character to video RAM
1953	0DBE	2A7111		LD	HL, (CURSOR)
1954	0DC1	7D		LD	A, L ; Character position
1955	0DC2	FE27		CP	39 ; Last character?
1956	0DC4	C2900E		JP	NZ, CURIGHT ; Bump cursor right
1957	0DC7	5C		LD	E, H
1958	0DC8	1600		LD	D, 0
1959	0DCA	217311		LD	HL, VIDMAP
1960	0DCD	19		ADD	HL, DE
1961	0DCE	7E		LD	A, (HL) ; Pick up line flag
1962	0DCF	B7		OR	A
1963	0DD0	C2900E		JP	NZ, CURIGHT ; Overflow?
1964	0DD3	23		INC	HL

1965	0DD4	3601		LD	(HL),1	; Set of low flag
1966	0DD6	23		INC	HL	
1967	0DD7	3600		LD	(HL),0	
1968	0DD9	C3390E		JP	SCROL2	
1969						
1970	0DDC	F5	CONTROL:	PUSH	AF	
1971	0DDD	C5		PUSH	BC	
1972	0DDE	D5		PUSH	DE	
1973	0DDF	E5		PUSH	HL	
1974	0DE0	47		LD	B,A	
1975	0DE1	E6F0		AND	OFOH	
1976	0DE3	FEC0		CP	OCOH	; Control character?
1977	0DE5	C2DE0F		JP	NZ,POPPER	
1978	0DE8	A8		XOR	B	; Isolate LSN
1979	0DE9	07		RLCA		
1980	0DEA	07		RLCA		; Multiply by 4
1981	0DEB	4F		LD	C,A	
1982	0DEC	0600		LD	B,0	
1983	0DEE	21F30D		LD	HL,JUMPTABLE	
1984	0DF1	09		ADD	HL,BC	; Where to go?
1985	0DF2	E9		JP	(HL)	; Control handler
1986						
1987	0DF3	C3320E	JUMPTABLE:	JP	SCROLL	
1988	0DF6	00		NOP		
1989	0DF7	C3740E		JP	LINEFEED	
1990	0DFA	00		NOP		
1991	0DFB	C3840E		JP	VTAB	
1992	0DFE	00		NOP		
1993	0DFF	C3900E		JP	CURIGHT	
1994	0E02	00		NOP		
1995	0E03	C3AE0E		JP	CURLEFT	
1996	0E06	00		NOP		
1997	0E07	C3BFOE		JP	HOME	
1998	0E0A	00		NOP		
1999	0E0B	C3C50E		JP	CLEAR	
2000	0E0E	00		NOP		
2001	0E0F	C3F80E		JP	DELETE	
2002	0E12	00		NOP		
2003	0E13	C3490F		JP	INSERT	
2004	0E16	00		NOP		
2005	0E17	C3E10E		JP	CAPITAL	
2006	0E1A	00		NOP		
2007	0E1B	C3EE0E		JP	SMALL	
2008	0E1E	00		NOP		
2009	0E1F	C3DE0F		JP	POPPER	
2010	0E22	00		NOP		
2011	0E23	C3DE0F		JP	POPPER	
2012	0E26	00		NOP		
2013	0E27	C38B0F		JP	CARRIAGE	
2014	0E2A	00		NOP		
2015	0E2B	C3DE0F		JP	POPPER	
2016	0E2E	00		NOP		
2017	0E2F	C3DE0F		JP	POPPER	
2018						
2019	0E32	CDA60D	SCROLL:	CALL	SYNC	
2020	0E35	AF		XOR	A	
2021	0E36	3203E0		LD	(PORT3),A	; Activate VGATE
2022	0E39	01C003	SCROL2:	LD	BC,40*24	; char to move
2023	0E3C	1100D0		LD	DE,VIDRAM	; Destination
2024	0E3F	2128D0		LD	HL,VIDRAM+40	; Source

2025	0E42	EDB0		LDIR		; Scroll up a line
2026	0E44	EB		EX	DE,HL	; HL points bottom line
2027	0E45	0628		LD	B,40	
2028	0E47	CDD80F		CALL	SCRUB	; Blank bottom line
2029	0E4A	011A00		LD	BC,26	; Line flags
2030	0E4D	117311		LD	DE,VIDMAP	; Point flags
2031	0E50	217311		LD	HL,VIDMAP	
2032	0E53	23		INC	HL	
2033	0E54	EDB0		LDIR		; And scroll line flags
2034	0E56	3600		LD	(HL),0	
2035	0E58	3A7311		LD	A,(VIDMAP)	
2036	0E5B	B7		OR	A	
2037	0E5C	C26A0E		JP	NZ,SCROL3	
2038	0E5F	CDA60D		CALL	SYNC	
2039	0E62	3E01		LD	A,1	
2040	0E64	3203E0		LD	(PORT3),A	; Restore Vsate
2041	0E67	C3DE0F		JP	POPPER	; Finish
2042	0E6A	2A7111	SCROL3:	LD	HL,(CURSOR)	
2043	0E6D	25		DEC	H	; Drop line count
2044	0E6E	227111		LD	(CURSOR),HL	
2045	0E71	C3390E		JP	SCROL2	; Carry on
2046						
2047	0E74	2A7111	LINEFEED:	LD	HL,(CURSOR)	
2048	0E77	7C		LD	A,H	; Pick up line count
2049	0E78	FE18		CP	24	; Last line?
2050	0E7A	CA320E		JP	Z,SCROLL	; Time to scroll
2051	0E7D	24		INC	H	; Bump line count
2052	0E7E	227111	EXIT:	LD	(CURSOR),HL	
2053	0E81	C3DE0F		JP	POPPER	
2054						
2055	0E84	2A7111	VTAB:	LD	HL,(CURSOR)	
2056	0E87	7C		LD	A,H	
2057	0E88	B7		OR	A	; Top line?
2058	0E89	CADE0F		JP	Z,POPPER	
2059	0E8C	25		DEC	H	; Drop line count
2060	0E8D	C37E0E		JP	EXIT	
2061						
2062	0E90	2A7111	CURIGHT:	LD	HL,(CURSOR)	
2063	0E93	7D		LD	A,L	; Pick up char count
2064	0E94	FE27		CP	39	; Past end of line?
2065	0E96	D29D0E		JP	NC,NEWLINE	
2066	0E99	2C		INC	L	; Bump character count
2067	0E9A	C37E0E		JP	EXIT	
2068						
2069	0E9D	2E00	NEWLINE:	LD	L,0	; Zero char count
2070	0E9F	24		INC	H	; Bump line count
2071	0EA0	7C		LD	A,H	
2072	0EA1	FE19		CP	25	; Overflowed?
2073	0EA3	DA7E0E		JP	C,EXIT	; OK so finish
2074	0EA6	2618		LD	H,24	; Set to last line
2075	0EA8	227111		LD	(CURSOR),HL	
2076	0EAB	C3320E		JP	SCROLL	
2077						
2078	0EAE	2A7111	CURLEFT:	LD	HL,(CURSOR)	
2079	0EB1	7D		LD	A,L	; Char count
2080	0EB2	B7		OR	A	; Start of line?
2081	0EB3	2804		JR	Z,CURL2	
2082	0EB5	2D		DEC	L	; Drop char count
2083	0EB6	C37E0E		JP	EXIT	
2084	0EB9	2E27	CURL2:	LD	L,39	; Last char

2085	OE8B	25		DEC	H		; Drop line
2086	OE8C	F27E0E		JP	P,EXIT		
2087	OE8F	210000	HOME:	LD	HL,0		; Set cursor to home
2088	OE82	C37E0E		JP	EXIT		
2089							
2090	OE85	CDA60D	CLEAR:	CALL	SYNC		; Wait
2091	OE88	0E19		LD	C,25		; 25 lines
2092	OECA	2100D0		LD	HL,VIDRAM		; Start
2093	OECD	0628	CLR2:	LD	B,40		; Chars per line
2094	OECF	CDD80F		CALL	SCRUB		; Blank a line
2095	OED2	0D		DEC	C		
2096	OED3	C2CD0E		JP	NZ,CLR2		; Keep going
2097	OED6	217311		LD	HL,VIDMAP		
2098	OED9	061B		LD	B,27		
2099	OEDB	CDD80F		CALL	SCRUB		; Scrub line flags
2100	OEDE	C3BF0E		JP	HOME		; Home-up
2101							
2102	OEEl	3E05	CAPITAL:	LD	A,5		
2103	OEEl	3203E0		LD	(PORT3),A		; Turn LED green!
2104	OEEl	3E00		LD	A,0		
2105	OEEl	327011	CAP2:	LD	(CASEFLAG),A		; Set case
2106	OEEl	C3DE0F		JP	POPPER		
2107							
2108	OEEl	3E04	SMALL:	LD	A,4		
2109	OEFO	3203E0		LD	(PORT3),A		; Turn LED red!
2110	OEFO	3E01		LD	A,1		
2111	OEFO	C3E80E		JP	CAP2		
2112							
2113	OEFO	2A7111	DELETE:	LD	HL,(CURSOR)		
2114	OEFO	7C		LD	A,H		
2115	OEFO	B5		OR	L		; At home?
2116	OEFO	CADE0F		JP	Z,POPPER		
2117	OF00	7D		LD	A,L		
2118	OF01	B7		OR	A		; Start of line?
2119	OF02	C21D0F		JP	NZ,DELT2		
2120	OF05	5C		LD	E,H		
2121	OF06	1600		LD	D,0		
2122	OF08	217311		LD	HL,VIDMAP		
2123	OF0B	19		ADD	HL,DE		
2124	OF0C	7E		LD	A,(HL)		; Pick line flag
2125	OF0D	B7		OR	A		
2126	OF0E	C21D0F		JP	NZ,DELT2		
2127	OF11	CDB10F		CALL	POSITION		; Find position
2128	OF14	CDA60D		CALL	SYNC		
2129	OF17	2B		DEC	HL		
2130	OF18	3600		LD	(HL),0		; Scrub it
2131	OF1A	C3AE0E		JP	CURLEFT		
2132	OF1D	2A7111	DELT2:	LD	HL,(CURSOR)		
2133	OF20	5C		LD	E,H		
2134	OF21	1C		INC	E		
2135	OF22	1600		LD	D,0		
2136	OF24	217311		LD	HL,VIDMAP		
2137	OF27	19		ADD	HL,DE		
2138	OF28	7E		LD	A,(HL)		
2139	OF29	47		LD	B,A		
2140	OF2A	B7		OR	A		
2141	OF2B	3E28		LD	A,40		
2142	OF2D	2802		JR	Z,DELT3		; Single line
2143	OF2F	3E50		LD	A,80		; Double line
2144	OF31	2A7111	DELT3:	LD	HL,(CURSOR)		

2145	OF34	95		SUB	L	
2146	OF35	4F		LD	C,A	
2147	OF36	0600		LD	B,0	; Characters to move
2148	OF38	CDB10F		CALL	POSITION	
2149	OF3B	E5		PUSH	HL	
2150	OF3C	D1		POP	DE	
2151	OF3D	1B		DEC	DE	; Destination
2152	OF3E	CDA60D		CALL	SYNC	
2153	OF41	EDB0		LDIR		; Move text left
2154	OF43	2B		DEC	HL	
2155	OF44	3600		LD	(HL),0	; Scrub char left at end
2156	OF46	C3AE0E		JP	CURLEFT	
2157						
2158	OF49	2A7111	INSERT:	LD	HL,(CURSOR)	
2159	OF4C	5C		LD	E,H	
2160	OF4D	1C		INC	E	
2161	OF4E	1600		LD	D,0	
2162	OF50	217311		LD	HL,VIDMAP	
2163	OF53	19		ADD	HL,DE	
2164	OF54	7E		LD	A,(HL)	
2165	OF55	B7		OR	A	
2166	OF56	0E00		LD	C,0	
2167	OF58	2A7111		LD	HL,(CURSOR)	
2168	OF5B	2E27		LD	L,39	
2169	OF5D	2802		JR	Z,INST2	
2170	OF5F	24		INC	H	
2171	OF60	0C		INC	C	
2172	OF61	CDB40F	INST2:	CALL	POS1	
2173	OF64	7E		LD	A,(HL)	
2174	OF65	B7		OR	A	
2175	OF66	C2DE0F		JP	NZ,POPPER	
2176	OF69	E5		PUSH	HL	
2177	OF6A	2A7111		LD	HL,(CURSOR)	
2178	OF6D	3E27		LD	A,39	
2179	OF6F	95		SUB	L	
2180	OF70	47		LD	B,A	
2181	OF71	79		LD	A,C	
2182	OF72	B7		OR	A	
2183	OF73	2804		JR	Z,INST3	
2184	OF75	3E28		LD	A,40	
2185	OF77	80		ADD	A,B	
2186	OF78	47		LD	B,A	
2187	OF79	D1	INST3:	POP	DE	
2188	OF7A	D5		PUSH	DE	
2189	OF7B	E1		POP	HL	
2190	OF7C	2B		DEC	HL	
2191	OF7D	CDA60D		CALL	SYNC	
2192	OF80	7E	INST4:	LD	A,(HL)	
2193	OF81	12		LD	(DE),A	
2194	OF82	3600		LD	(HL),0	
2195	OF84	2B		DEC	HL	
2196	OF85	1B		DEC	DE	
2197	OF86	10F8		DJNZ	INST4	
2198	OF88	C3DE0F		JP	POPPER	
2199						
2200	OF8B	2A7111	CARRIAGE:	LD	HL,(CURSOR)	
2201	OF8E	5C		LD	E,H	
2202	OF8F	1C		INC	E	
2203	OF90	1600		LD	D,0	
2204	OF92	217311		LD	HL,VIDMAP	

2205	0F95	19		ADD	HL, DE	
2206	0F96	7E		LD	A, (HL)	
2207	0F97	B7		OR	A	
2208	0F98	2A7111		LD	HL, (CURSOR)	
2209	0F9B	CA9D0E		JP	Z, NEWLINE	
2210	0F9E	2E00		LD	L, 0	
2211	0FA0	7C		LD	A, H	
2212	0FA1	FE17		CP	23	
2213	0FA3	2805		JR	Z, CAR2	
2214	0FA5	24		INC	H	
2215	0FA6	24		INC	H	
2216	0FA7	C37E0E		JP	EXIT	
2217	0FAA	24	CAR2:	INC	H	
2218	0FAB	227111		LD	(CURSOR), HL	
2219	0FAE	C3320E		JP	SCROLL	
2220						
2221	0FB1	2A7111	POSITION:	LD	HL, (CURSOR)	
2222	0FB4	C5	POS1:	PUSH	BC	
2223	0FB5	D5		PUSH	DE	
2224	0FB6	E5		PUSH	HL	
2225	0FB7	C1		POP	BC	; BC = Line, char counter
2226	0FB8	112800		LD	DE, 40	
2227	0FB8	21D8CF		LD	HL, VIDRAM-40	
2228	0FBE	19	POS2:	ADD	HL, DE	; Add a line
2229	0FBF	05		DEC	B	
2230	0FC0	F2BE0F		JP	P, POS2	; More lines?
2231	0FC3	0600		LD	B, 0	
2232	0FC5	09		ADD	HL, BC	; Add in char count
2233	0FC6	D1		POP	DE	
2234	0FC7	C1		POP	BC	
2235	0FC8	C9		RET		; HL points to video pos
2236						
2237	0FC9	2A03E0	INIT8255:	LD	HL, (PORT3)	; 8255 PPI
2238	0FCC	368A		LD	(HL), 8AH	; A=OUT, B=IN, C=I/O
2239	0FCE	3607		LD	(HL), 7	; C3 HI (MOTOR OFF)
2240	0FD0	3605		LD	(HL), 5	; C2 HI (LED green)
2241	0FD2	3E01		LD	A, 1	
2242	0FD4	3203E0		LD	(PORT3), A	; C0 HI (VGATE on)
2243	0FD7	C9		RET		
2244						
2245	0FD8	AF	SCRUB:	XOR	A	
2246	0FD9	77	SCR2:	LD	(HL), A	; Zero whatever
2247	0FDA	23		INC	HL	
2248	0FDB	10FC		DJNZ	SCR2	; Keep going
2249	0FDD	C9		RET		
2250						
2251	0FDE	E1	POPPER:	POP	HL	
2252	0FDF	D1	POP2:	POP	DE	
2253	0FE0	C1		POP	BC	
2254	0FE1	F1		POP	AF	
2255	0FE2	C9		RET		
2256						
2257				END		